E5 ENCLAVE INCORPORATED

The Measure of the Wound

A Sovereign Empirical Record of Black American Structural Distress

1991 – 2024

Israel Lee Armstead

E5 Enclave Incorporated · Liberty City, Miami, Florida

CORRECTED PRINT EDITION · BLACK PAPER V1.4 · SUBMISSION EDITION

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Underlying data. All source data is public and openly licensed. Layer 1 raw evidence: `IAMGODIAM/bdi-raw-data-vault`. Layer 2 synthesized instrument: `IAMGODIAM/bdi-sovereign-dataset`, sealed on Base Mainnet, ExodusV4 token #2. Layer 3 place-level application: `IAMGODIAM/farmblock-data` and `IAMGODIAM/farmblock-dataset`.

On this edition. Every derived statistic was recomputed from the raw source series rather than carried forward from prior drafts, and flagged figures were re-verified against live federal sources in August 2026. Twenty-seven public claims were put through evidentiary triage; ten further arithmetic errors were found by recomputation; three canonical counts in the project's own governance documents were found stale. This edition then incorporates a second wave of corrections arising from an independent review of the v1.1 print edition (September 1, 2026), which identified a defective price basis in the Chapter 5 wealth series and an unverified denominator in the incarceration series. Both are corrected; several claims are withdrawn. All corrections are enumerated in Appendix H, sections A, B and B2. The Submission Edition (v1.3) added an abstract, an author of record, an AI-assistance disclosure, keywords and JEL codes, and a consolidated References section. This edition (v1.4) states the FarmBlock reproducibility limit in Chapter 7 where the index is introduced rather than in the appendix alone, and rewrites the AI disclosure to describe the division of labor exactly rather than approximately. Neither edition changes any finding.

Nil satis nisi optimum.

DEDICATION

I dedicate this work to the countless love warriors, wounded healers and freedom fighters who trod this path before me, in this moment in the movement for Black American Lives, Liberty, and Economic Parity.

The work, the scholarship and the brotherhood of Professor Cornel “Brother” West provided invaluable wisdom and a wellspring of inspiration in this endeavor.

To my late grand-uncle, Ralph C. McCartney — thank you for your instruction and your guidance. I carry forth the mantle of your work and your legacy in the struggle for Black Upliftment, and I carry it proudly.

To my brilliant wife and willing editor, Elvia G. Brazil-Armstead — thank you for your heart, your hand and your pen.

To the descendants of chattel-enslaved persons — the village, our community — may the record set down here bring nearer the repair that is owed you for the vestiges and the harms you have so long endured.

I love you, All.

— Israel Lee Armstead

LIBERTY CITY, MIAMI

ABSTRACT

This paper assembles an eight-pillar empirical record of Black American structural disparity across the era of formal legal equality, 1991–2024, nested within federal series that reach to 1900 (health), 1925 (criminal justice), 1940 (housing), 1964 (political participation), 1972 (labor) and 1514 (the historical architecture). Drawing on approximately 14,811 raw observations from the Bureau of Labor Statistics, the Census Bureau, the National Center for Health Statistics, the Bureau of Justice Statistics, the Federal Reserve, the National Center for Education Statistics, the Consumer Financial Protection Bureau, the Department of Agriculture and the Slave Voyages Database, synthesized into 1,574 verified observations and applied to 15,507 census tracts in 49 cities, it documents a consistent pattern: absolute conditions improve while the ratio between Black and white outcomes holds or widens. In constant 2022 dollars, Black median family wealth rose 388 percent while the absolute wealth gap reached its widest point in the survey's history. The Black/white unemployment ratio has never inverted in fifty-four years. The maternal mortality ratio is higher in 2022 (2.61) than in 1930 (1.48). The imprisonment ratio moved 0.14 points across ninety-seven years. The homeownership gap is wider than when the Fair Housing Act was signed. The paper contributes a place-level composite index, prints its full corrections ledger as an appendix, publishes an independent verification review and the author's reply alongside the release, and places all data and code in the public domain under CC0.

Keywords: racial inequality; wealth gap; structural racism; Black Americans; longitudinal federal data; composite index; maternal mortality; incarceration; homeownership; open data

JEL codes: J15, D63, I14, I24, K42, R21, N32

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Disclosure of AI assistance. This work used generative AI. Here is exactly where. The author wrote the prose. A human editor, Elvia G. Brazil-Armstead, read the draft and returned handwritten notes; Claude (Anthropic) then applied line edits against those notes and the author's instructions. Claude also did the arithmetic — pulling the series from federal sources the author had chosen, recomputing every derived statistic, and compiling the corrections ledger — and it built the typesetting pipeline. It originated no research question, selected no source, drew no conclusion and made no claim. The author specified each computation and checked each figure it returned against the primary series; where the two disagreed, the correction is printed in Appendix H. Manus AI performed an independent verification review of the v1.1 edition; that review and the author's reply are published in the project repository. No AI system is an author of this work and none is credited as one. Every numeric claim traces to a named federal or archival source in the References and in Appendix A.

Data and code availability. All source data, manuscript sources, the build pipeline, the recomputation log and the corrections ledger are public under CC0 1.0 in the publication package at github.com/IAMGODIAM/measure-of-the-wound, in the working repository at github.com/IAMGODIAM/bdi-black-paper, and in the four data repositories cited in Appendix F. The package carries a frozen snapshot of every source file the tables are computed from, with a SHA-256 manifest, and rebuilds the paper byte-for-byte from a pinned container. The FarmBlock Distress Index outputs are published but not yet independently reproducible from the released package; see Appendix E.4a.

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PREFACE

We publish this paper in 2026 because the record demands it.

The empirical case for the structural dispossession of Black Americans has never been stronger. The tools to make that case have never been more accessible. And the institutional willingness to hold that case in front of the people with power to act on it has, in this moment, rarely been more fragile. Datasets documenting racial inequality are being contested and removed from federal databases — the EPA's own environmental justice screening tool was taken down from EPA.gov on February 5, 2025, and survives publicly only because an independent archive preserved it. Research programs that have tracked Black health, wealth and housing outcomes for decades are being defunded. The language of structural racism — a precise, evidence-based analytical category — is being litigated out of public discourse at the same moment the disparities it describes are widening.

We do not read this moment as uniquely terrible in the history of this country. The suppression of evidence about Black American life is not new. What is new is that the tools now exist to produce a sovereign record — one belonging to no government, no foundation, no university and no political cycle. That record cannot be revised under pressure. It cannot be taken behind a paywall. It cannot be defunded.

This paper is that record.

What did not exist before this paper is specific: a single, freely available, empirically comprehensive instrument measuring all major dimensions of Black American structural distress — economic, health, criminal justice, education, housing, political, environmental and historical — in one place, dual-source verified, licensed for unrestricted public use.

The federal government produces fragments of this record. The Bureau of Labor Statistics measures unemployment. The National Center for Health Statistics tracks maternal mortality. The Census Bureau publishes wealth, income and poverty data. The Bureau of Justice Statistics documents incarceration. Each source is rigorous. None of them synthesize. None are designed to tell a unified story across eight dimensions of the same life.

Academic institutions have produced that synthesis — in peer-reviewed journals accessible to researchers with institutional affiliations and subscription budgets. The communities at the center of the data have not been among the primary beneficiaries of that work.

This paper corrects that. The dataset underlying it is CC0, public domain. Every table, every trend line, every compound score, every city in the analysis belongs to anyone who can use it: the community organizer building a grant application, the attorney preparing a fair housing complaint, the journalist writing about the Delta, the student at an HBCU whose library cannot afford the journal that published the study she needs. There is no request form. There is no fee. There are no restrictions. It is yours.

E5 Enclave Incorporated is a Black-led nonprofit headquartered in Liberty City, Miami, Florida. Liberty City is not incidental to this paper. It is one of the communities the data documents — a neighborhood redlined in the 1930s, displaced by highway construction in the 1960s, the site of the 1980 rebellion, and still, in 2026, carrying a poverty rate, food access gap and homeownership deficit that place it inside the compound distress geography this paper maps.

E5 Enclave was established to build sovereign infrastructure — institutions that do not require the permission of their opposition to function, data that does not disappear when the political wind shifts, technology that transfers analytical power to communities rather than extracting intelligence from them. This paper is that mission expressed as an empirical act.

The dataset behind it was produced without government funding, without foundation grants and without academic institutional support. That independence was not circumstantial. It was a design requirement. A record this important cannot be produced under conditions that create competing obligations to its accuracy.

This paper is addressed to everyone who needs it.

To the funder whose portfolio has been pressured to retreat from racial equity work: this is the empirical record of what that retreat costs, measured in years of life, percentage points and dollars.

To the congressional staffer preparing testimony on federal housing enforcement: Chapter 5 documents that fifty-three percent of public housing authorities did not file the Section 3 reports required by a 1968 statute — which means that for more than half the program, whether the law was honored is not merely unproven but unknowable.

To the attorney litigating an environmental justice case in the Louisiana petrochemical corridor: Chapter 6 documents 150 industrial facilities across an 85-mile stretch, and specifies the geography precisely — fence-line census tracts in St. James Parish Districts 4 and 5 are 65 to 94 percent Black, against a parish-wide share of approximately 44 percent. The distinction is the case.

To the community organizer in Humphreys County, Mississippi — FarmBlock distress score 87.25, the highest in the corpus; no acute-care hospital since 2013; hypertension prevalence of 51.7 percent; 55 percent of children below the poverty line: you have the data now, with its full decomposition and its margins of error printed alongside it. It belongs to you.

To the researcher at Howard, Spelman, Morehouse, or any institution serving the communities this paper documents: build on it, challenge it, extend it. The methodology is transparent. The sources are cited. The errors we found in our own drafts are printed in Appendix H. The license is CC0. That is the invitation.

The tradition this paper enters is long.

Ida B. Wells counted the dead and published the names when no institution would. W.E.B. Du Bois applied the full force of social science methodology to Black American life when the academy insisted the question was not worth asking. Ella Baker built organizing infrastructure requiring no charismatic center, because she understood that sovereign capacity cannot live in one person. James Baldwin refused the distance of academic prose because the subject was too important for comfort. Cornel West insisted that the empirical and the moral must travel together — that data without witness is arithmetic, and witness without data is sentiment.

And there is Ralph C. McCartney of Overtown, Miami — community organizer, oral historian, veteran of decades of displacement resistance, honored on the floor of the United States Congress by Congresswoman Carrie Meek on February 1, 1994. In 1997 he recorded his testimony about what the construction of I-95 and I-395 did to Overtown: a neighborhood where "never did a hungry person cross that door and leave out that same way," where teachers and principals were "like extended members of the family." The expressways bisected that community into four disconnected fragments. The property was taken. The compensation was inadequate. The plans, McCartney said, had been on the books for fifty years: "*They know everything there is to know about us. The only problem we have is what they decide to do with us.*"

His testimony is primary source evidence. He was also my grand-uncle. The data in this paper is, among other things, the empirical record at national scale of what happened to the community he spent his life defending.

This paper is built in that tradition. It is empirical and it is moral. The two are not in tension. In the face of what this data shows, precision and witness are the same act.

A note on what this edition is.

This is the corrected print edition. Between the working drafts and this printing, every public claim was put through an evidentiary triage, every derived statistic was recomputed from the raw source series rather than carried forward, and the flagged figures were re-verified against the live federal sources in August 2026.

Twenty-seven claims were triaged; eleven conflicted with their sources and four had been derived internally without published methodology. Ten additional arithmetic and transcription errors were found in the drafts. Three canonical counts in our own governance documents were stale. The highest-scoring tract in our published index turned out to be an imputation artifact.

Then the corrected edition was itself reviewed, independently, by a reader with no access to our repositories — and that review found the most serious error in the paper: the wealth table in Chapter 5 spliced a constant-dollar endpoint onto a nominal history, without saying so, and its 1989 figures could not be reconciled to the Federal Reserve's published race-specific medians under either basis. The table has been rebuilt from the Federal Reserve's own series. A headline claim about the wealth gap tripling, and a projection about parity, are withdrawn. So is an overstatement about the scope of the incarceration ratio.

Every one of these is corrected in the text and enumerated in Appendix H, with the original claim, the verified value and the reason for the change.

That second wave matters more than the first. Our own audit checked whether our arithmetic followed from our data; it never asked whether our data was coherent. An internal consistency check cannot catch an incoherent input. It took an outside reader to find that, which is the entire argument for publishing under a licence that invites one.

We print the errata because a sovereign record that conceals its corrections is not sovereign — it is promotional. The dataset is CC0 and the sources are public precisely so that this paper can be checked. It would be incoherent to invite that scrutiny and then hide what our own scrutiny found.

E5 Enclave Incorporated authors this paper as an institutional act of witness. The data is the contribution. The communities it documents are the authority.

INTRODUCTION

THE WOUND

> "I cried because there was one white man who had the power to say to another white man
> *— finish wiping out this Black history.*" — Ralph McCartney, oral history, August 14, 1997, Black Archives of South Florida

The wound is not a metaphor. It is measurable. It has a geography, a timeline, a set of federal series numbers, and a spreadsheet. This paper measures it.

That sentence is the whole argument, and everything that follows is its documentation. What has been written about Black American structural distress has most often been written in one of two registers — the moral, which is true and unquantified, or the technical, which is quantified and fragmented across a dozen federal agencies that do not speak to one another. This paper refuses the choice. It holds both.

The Window, and What Is Nested Inside It

The core empirical window of this paper is 1991 to 2024 — thirty-three years, chosen deliberately. It is the era of formal legal equality. The Civil Rights Act is twenty-seven years old when the window opens. The Voting Rights Act is fully extended. The Fair Housing Act has been law for a generation. Whatever is measured inside this window cannot be attributed to explicit legal segregation, because explicit legal segregation had already been abolished. The disparities documented here exist *within* the framework of equal protection, produced by systems that operate lawfully.

That is the point. A gap that closes when the law changes is a gap the law was causing. A gap that does not close when the law changes is a gap something else is causing — and the something else is what this paper measures.

But the thirty-three-year window does not explain itself. So where federal data permits, the series run longer, and each longer series is nested inside the argument rather than appended to it:

- **Health to 1900** — 122 years of life expectancy by race, and maternal mortality to 1915.
- **Criminal justice to 1925** — 97 years of incarceration rates by race.
- **Housing to 1940** — 82 years of decennial homeownership.
- **Labor to 1972** — 54 years of monthly BLS unemployment, the longest continuous race-disaggregated federal labor series in American public data.
- **Political participation to 1964** — 56 years of Census voting supplements.
- **The historical architecture to 1514** — 352 years of documented transatlantic voyages.

These extensions are not context. They are causation — the documented mechanism that produced the conditions the thirty-three-year window measures. A wealth gap measured in 2022 is the present-tense reading of an instrument that was calibrated in 1866, and again in 1934, and again in 1968.

What Has Been Measured

Three layers, each publicly auditable, each committed before the analysis that used it:

~14,811 raw federal observations, pulled from seventeen source files across BLS, the Census Bureau, NCHS, BJS, the Federal Reserve, USDA, NCES, CFPB and the Slave Voyages Database, and committed unmodified to a public repository before any scoring occurred.

1,574 verified empirical observations, synthesized into an eight-pillar instrument spanning the empirical window, dual-source confirmed, and sealed on a public blockchain.

15,507 census tracts, scored across forty-nine cities in seventeen states, where the national pattern acquires a street address.

The middle figure was once published as 1,855. It was wrong — the count had included JSON metadata keys as though they were observations. The correction to 1,574 is documented in the counting specification, disclosed in Chapter 3, and printed in Appendix B. It is recorded here, in the introduction, rather than buried in a methods note, because a record that corrects itself in public is the only kind worth trusting.

The Compound Catastrophe Zones

Eight pillars measured separately produce eight findings. Measured together, they produce something different: a small number of places where every pillar has failed at once, and where the failures reinforce each other faster than any single-pillar intervention can address them.

The instrument's five highest-scoring counties are Humphreys, Claiborne, Sunflower and Amite in Mississippi, and Alexander County, Illinois. These are not the poorest places in America by any single measure. They are the places where poverty, health burden, housing abandonment, digital exclusion and structural exposure arrive simultaneously and compound.

Readers of earlier drafts will notice that this is not the list those drafts named. They named Humphreys, Detroit, East St. Louis, Claiborne and the Louisiana petrochemical corridor — a selection made for documentary depth rather than by score. Detroit ranks eleventh. East St. Louis is not in the published corpus at all. That list is withdrawn, and Chapter 7 sets out both the corrected ranking and the reason a composite index must name the places its own composite produced.

What This Paper Is, and Is Not

It is **not a policy brief**, though policy follows directly from every chapter. It is **not an academic monograph**, though the evidentiary standard is meant to survive the same scrutiny. It is **not a foundation report**, because no funder shaped its conclusions — there was no funder. It is **not a protest document**, though the findings are damning enough to serve as one.

It is a sovereign empirical record: eight-dimensional, dual-source verified, released into the public domain under CC0, sealed on-chain against revision, and produced by an organization headquartered inside the geography it measures.

And it is a corrected record. Twenty-seven public claims were put through an evidentiary triage before this edition was set in type. Eleven were found to conflict with their sources. Four were internally derived without published methodology. Two carried stale citations. Ten independent recomputation errors were found in the drafts themselves. Every one of them is disclosed, with the corrected figure, in Appendix H.

That appendix is not an embarrassment. It is the credential. A record whose numbers have never been challenged is a record no one has checked.

The Reader's Obligation

The data settles the factual question. Whether the wound is real, whether it is measurable, whether it has persisted across the full period of formal legal equality — these are no longer open questions, and this paper is built so that anyone who wishes to verify that can do so without permission, payment, or institutional affiliation.

The moral question is untouched by any of it. It belongs to whoever is holding this paper.

Turn the page.

PART ONE

The Tradition and the Gap

CHAPTER 1

From Turner to Baker to West: The Unbroken Line

Every generation of Black liberation has produced its instrument. Not a slogan. Not a symbol. An instrument — something that functions, that carries weight, that can be picked up and used by the next person in line. The tradition does not run from a single source. It is a braid: methodology and resistance, testimony and infrastructure, empirical rigor and moral urgency, each strand necessary, none sufficient alone. This paper is the latest instrument in that braid. It did not arrive without ancestry. Nothing sovereign ever does.

The Body as First Instrument

In August of 1831, Nat Turner led the only large-scale, sustained slave rebellion in American history. He was a literate man, a preacher, and a tactician. He understood that the law did not acknowledge the value of his life, and that waiting for the law to change was itself a choice — with consequences. What Turner did was declare, in the only language the system was prepared to hear, that Black life carried a sovereignty the law refused to name. He is not honored here for the deaths that followed his rebellion — of the enslaved who were executed, of the white Southerners killed, of the hundreds of Black people massacred by militia in the weeks of retribution that followed. He is named because the act of refusal was prior to every instrument that came after it. Before the narrative, before the database, before the policy platform — there was a man who looked at the architecture of his captivity and said: this has a cost you have not yet been asked to pay. Turner did not survive. The instrument he built did not survive. What survived was the proof that the architecture could be refused.

Infrastructure as Liberation

Harriet Tubman built a system. Not a symbol — a logistics operation. Between 1849 and 1860, she made nineteen return trips into slave territory and liberated approximately seventy people. Zero were captured. The Underground Railroad — a term applied retroactively to a network of routes, safe houses, contacts, and protocols that Tubman and others developed and maintained — was the first model of what this organization calls sovereign infrastructure: decentralized, requiring no institutional permission, built to operate regardless of what the law said.

The methodological lesson is not courage. It is architecture. Tubman did not build a movement that depended on a center. She built a network that could survive the loss of any node. The people she liberated did not follow a leader. They followed a system. When she said she never lost a passenger, she was not describing inspiration. She was describing operational discipline.

Every subsequent instrument in this tradition has faced the same design question Tubman solved: how do you build something that cannot be shut down? The answer she gave in 1850 is the same answer this paper gives in 2026 — you make it sovereign, you make it distributed, and you make it free to anyone who needs it.

Testimony as Data

Frederick Douglass's Narrative of the Life of Frederick Douglass, an American Slave, published in 1845, was an act of empirical self-documentation before the field of empirical social science existed. Douglass understood that the argument against slavery was not primarily moral — the moral argument had been made, repeatedly, and ignored. The argument that could not be dismissed was evidentiary: here is a man, here is what was done to him, here is the date, here is the place, here is the name of the person who did it. The Narrative is a dataset. It has subjects, conditions, outcomes, and sources. Ida B. Wells extended this method into journalism and turned it into political power. Beginning in 1892, following the lynching of three of her close friends in Memphis, Wells began systematically collecting records — names, dates, locations, stated justifications, actual circumstances — of every lynching she could document. She published them when no institution would. She cross-referenced the stated justifications against the actual evidence and demonstrated, with data, that the legal pretexts were fabrications. Her 1895 report *A Red Record* is the first Black investigative dataset in American history. Wells did not document suffering in order to produce sympathy. She documented it in order to produce accountability. The tradition established by Douglass and Wells — testimony transformed into irrefutable evidence — is the direct ancestor of the dual-source verification methodology that underlies every data point in this paper.

Rigor as Weapon

W.E.B. Du Bois published *The Philadelphia Negro* in 1899. It was the first systematic sociological study of a Black American urban community, and Du Bois produced it at a moment when the academy had not only declined to study Black American life but had constructed elaborate theoretical frameworks to justify that neglect. He used the tools of empirical social science — household surveys, economic analysis, historical records, occupational mapping — to document what the conditions of Black Philadelphia actually were, as opposed to what the dominant narrative claimed them to be.

The insight that drove the methodology was precise: the system producing inequality had a vested interest in misrepresenting the results of its own operation. Independent empirical documentation was therefore not a scholarly exercise. It was a counter-intelligence operation. Du Bois gave the tradition its methodology — the insistence that the argument must be grounded in data rigorous enough that it cannot be dismissed on methodological grounds, only on political ones.

When he wrote, in 1903, that "the problem of the twentieth century is the problem of the color line," he was not making a prediction. He was reading the empirical record he had already assembled. The data in this paper extends that reading by a century. The color line is still there. Now we have thirty-three years of federal data at the core, and a century or more in several pillars, to show exactly where it runs.

Architecture Without Leaders

Ella Baker spent forty years building organizing infrastructure that deliberately refused the model of charismatic leadership. As a field organizer for the NAACP in the 1940s, as the executive director of the Southern Christian Leadership Conference in the late 1950s, and as the founding mentor of the Student Nonviolent Coordinating Committee in 1960, Baker's consistent argument was the same: movements that depend on a singular leader are movements that can be stopped by stopping one person. The FBI understood this. COINTELPRO was, among other things, a systematic test of her thesis — and the assassinations, imprisonments, and manufactured scandals that dismantled the leadership of the Black liberation movement in the late 1960s confirmed it.

Baker's model was the inverse: develop local leadership capacity, build horizontal networks, make every node sufficient to the task. "Strong people don't need strong leaders." The data in this paper is designed as a Baker instrument. It requires no institutional intermediary to use. It names no individual author. It transfers analytical capacity directly to whoever holds it. The organization that produced it does not need credit. The data needs to be used.

The Current Line

Malcolm X gave the tradition its declaration of sovereignty over the definition of legitimate response — the insistence that Black people had the right to determine, for themselves, what constituted an appropriate answer to the conditions being imposed on them. It was not a call to violence. It was a refusal of the frame in which Black response to violence was the only thing being regulated.

James Baldwin gave the tradition its voice. Not the voice of documentation, not the voice of policy — the voice of witness. Baldwin understood that academic distance from the data was a moral failure, not a methodological virtue. The writer's obligation was to make the reader feel the weight of what the numbers meant in the lives of actual people. Every chapter in this paper attempts to honor that obligation: the data is precise, and the numbers represent people who had names and addresses and children and ambitions.

Cornel West gave the tradition its philosophical insistence for the modern era: that the empirical and the moral cannot be separated without damage to both. Data without witness is arithmetic. Witness without data is sentiment. West demanded both simultaneously, and the discipline of holding them together — refusing the comfort of pure analysis and the comfort of pure feeling — is the operating standard of this paper.

The Movement for Black Lives, launched in the wake of the police killing of Trayvon Martin in 2012 and formalized as a policy platform by 2016, produced the most comprehensive Black liberation policy framework in the modern era — a document that mapped demands across criminal justice, economic justice, reparations, political power, and community control. Dr. Melina Abdullah, co-founder of the Los Angeles chapter of Black Lives Matter and one of the movement's most disciplined organizers, embodied the Baker principle within the modern context: community accountability over celebrity leadership, local organizing as the primary unit of change, the street and the policy room as connected arenas rather than competing ones.

And Ralph McCartney of Overtown, Miami — community organizer, veteran, oral historian, honored on the floor of the United States Congress by Congresswoman Carrie Meek for a life spent in service to his community — gives the tradition its local fidelity. In 1997, McCartney sat for an oral history interview that would be archived by the Black Archives of South Florida and cited decades later in academic literature. He described what was done to Overtown — the expressways, the displacement, the calculated destruction of a self-sufficient community — with the precision of a man who had watched it happen from inside the geography. "They have their plans set decades and scores in advance," he said. "50 to 60 years ago these plans were on the books and it's coming to pass every day."

He was not speaking metaphorically. The HOLC redlining maps confirm it. The Urban Renewal displacement records confirm it. The city commission records he referenced confirm it. The data in this paper confirms it at national scale, across eight dimensions — and across series that reach back to 1900 in health, 1925 in criminal justice, 1940 in housing and 1514 in the historical architecture.

McCartney's grand-nephew, Israel Armstead, is the Executive Director of E5 Enclave Incorporated, headquartered in Liberty City — the community McCartney helped shape and refused to abandon. The line runs unbroken from the McCartney Palace on Northwest 15th Street to the organization that produced this paper.

Each built on the last. Each was met with resistance, then partial acknowledgment, then systematic absorption that diluted the demand. This paper is next in that line.

CHAPTER 2 — The Data Gap: What Has Never Been Built

The tradition documented in Chapter 1 has never lacked for moral clarity. What it has lacked, at every stage, is a single instrument that could make the full scope of the wound legible in one place — empirically comprehensive, structurally organized, freely available, requiring no institutional affiliation to access, produced by and accountable to the communities at the center of the data. That instrument did not exist before this paper. This chapter explains precisely what was missing and why.

The Government Record: Rigorous and Fragmented

The federal government has assembled the raw material. The Bureau of Labor Statistics tracks Black unemployment every month. The Centers for Disease Control measures maternal mortality by race. The Census Bureau produces annual wealth, poverty, and homeownership data through the Survey of Consumer Finances and the American Community Survey. The Bureau of Justice Statistics documents incarceration rates by race. The National Assessment of Educational Progress measures academic outcomes by demographic group. The Home Mortgage Disclosure Act requires lenders to report mortgage approvals and denials by race. The EPA's EJScreen maps environmental burden by census tract. The National Archives holds the Freedmen's Bureau records. The Slave Voyages Database documents 12,521,337 Africans forced onto slave ships and 10,702,656 who survived to reach the Americas — a precision that did not exist in any single source twenty years ago.

Each of these instruments is rigorous within its lane. None of them synthesize. None of them are designed to tell a unified story across eight dimensions of the same life in the same community. The unemployment data does not speak to the incarceration data. The housing data does not speak to the environmental data. The historical record does not speak to the present-tense disparity data. They exist in separate agencies, separate databases, separate annual reports — each requiring a different data literacy, a different institutional access point, a different vocabulary.

The community that is the subject of this data is not the primary audience it was designed to serve.

The Academic Record: Synthesized and Inaccessible

Academic researchers have done the synthesis. The scholarship on racial wealth gaps, carceral expansion, environmental racism, educational opportunity gaps, and housing discrimination is extensive, rigorous, and — behind institutional paywalls that cost forty dollars per article without a university subscription — largely unavailable to the communities and advocates who need it most.

The peer-reviewed literature has done its work. The data exists in journals. It has been reviewed. It has been cited by policymakers who have used it to produce commission reports that confirm the disparity, recommend reforms, and return to the same shelves as the studies that preceded them.

The academic tradition has not failed to produce knowledge. It has structured that knowledge in a way that does not transfer power to the communities the knowledge is about. This is not a critique of individual researchers. It is a structural observation about who the instrument of academic publication is designed to serve. A community organizer in Humphreys County, Mississippi — no university affiliation, no library subscription, a FarmBlock distress score of 87.25 — the highest in the scored corpus — does not have access to the literature that documents the conditions she lives inside. That is a design problem, not an accident.

The Advocacy Record: Urgent and Single-Pillar

The Movement for Black Lives' *A Vision for Black Lives* policy platform (2016) is the most comprehensive modern articulation of Black liberation demands in the policy register. The NAACP publishes annual criminal justice reporting. The National Fair Housing Alliance tracks mortgage discrimination and fair housing enforcement. The National Urban League's *State of Black America* has appeared annually since 1976. The Brookings Institution, the Economic Policy Institute, and the Urban Institute have produced significant work on racial wealth gaps, housing, and labor markets.

Each is rigorous. None of them are what this paper is.

The advocacy reports are designed for specific policy campaigns — they build the case for a particular intervention in a particular pillar. The academic reports are designed for peer review — they establish findings within a methodological framework. Neither is designed as a sovereign, eight-dimensional, dual-source verified, CC0-licensed, on-chain public instrument that belongs to anyone who can use it.

The gap is not one of effort or intelligence or commitment. Every organization named above has produced work of consequence. The gap is structural: no single instrument has combined all eight dimensions of structural distress, dual-source verified, into a public document owned by no institution, paywalled by no gatekeeper, and produced by a Black-led organization without government funding or foundation conditions.

The Gap, Precisely Stated

What did not exist before this paper is specific. Not a synthesis — those exist. Not a dataset — the federal government produces those. Not an advocacy platform — those exist and have shaped policy. What did not exist was this combination:

Eight-dimensional. Economic, health, criminal justice, education, housing, political, environmental, and historical — not as separate reports but as a single instrument in which the relationships between pillars are visible and the cumulative weight can be measured as a compound score.

Longitudinally unprecedented. The federal government has collected racial data for over a century — but in separate agencies, across incompatible series, in publications never aggregated into a single instrument. This dataset integrates BLS unemployment to 1972 (fifty-four years), NCHS health data to 1900 (one hundred and twenty-two years), BJS incarceration to 1925 (ninety-seven years), Census homeownership to 1940 (eighty-two years) and Slave Voyages records to 1514. No existing public document assembles these series in one instrument, dual-source verified and CC0 licensed.

Dual-source verified. Every data point confirmed against at least two independent sources. Where sources disagree, the more conservative estimate is used and the discrepancy is documented. This is a higher verification standard than most advocacy reports apply and matches the standard of peer-reviewed academic work — without the paywall.

CC0. Not "free to read." Not "open access." CC0 — public domain. No license required. No attribution required. No restrictions. Copy it. Translate it. Build on it. Use it in litigation, in testimony, in a grant application, in a classroom, in a community meeting in Humphreys County at midnight with no internet access and a printed copy.

On-chain. The dataset is sealed on the Base Mainnet blockchain. It cannot be revised under political pressure. It cannot be quietly altered. The record exists independent of any server, any institution, any administration.

Produced without conditions. E5 Enclave Incorporated built this dataset without government funding, without foundation grants, without academic institutional support. Every condition of funding is a potential condition of finding. This dataset has no funders to satisfy. It has one obligation: accuracy.

Black-led. The organization that produced this paper is headquartered in Liberty City, Miami, Florida — inside the compound distress geography it documents. The people who built this dataset live in the data. That is not a credential. It is an accountability structure.

That combination — not any single element — is the novelty this paper represents. The BDI does not replace what came before. It is the synthesis layer that makes all of it more useful to the people who most need it: the attorney preparing a fair housing complaint, the organizer building the case for a city council member, the congressional staffer drafting testimony on maternal mortality, the researcher at an HBCU whose library cannot afford the journal that published the study she needs, the community in Humphreys County that has been waiting for someone to put all of it in one place and hand it over.

Here it is.

PART TWO

The Data Architecture

CHAPTER 3

THE STACK: HOW THE DATA WAS BUILT

3.1 Research Design Overview

This research employs a mixed-method observational design combining longitudinal secondary data analysis with cross-sectional composite index construction. The project produces two distinct but architecturally linked instruments:

Product A — The Black Distress Index (BDI): A national longitudinal structural record documenting eight dimensions of Black community disparity from 1991 through 2024, with historical framing extending to 1514. The BDI operates as a forensic documentation system — it does not model cause and effect. It records what happened and when, across verifiable federal and institutional data series.

Product B — The FarmBlock Food Distress Index (FDI): A place-based intervention prioritization instrument scoring structural co-occurrence at the census tract level (15,507 tracts, 49 cities) and at the county level (24-county published pilot; approximately 3,144 counties in estimated full coverage). The FDI is a correlation instrument. It identifies communities where structural burdens cluster. It does not assert that any one dimension causes another.

The two instruments share a source architecture and governance protocol but serve different analytical functions. The BDI answers the question: *What has the structural record shown across time?* The FDI answers: *Where do these conditions concentrate now, and in what combination?*

Neither instrument makes causal claims. Both are designed for use in grant applications, policy testimony, litigation support, community organizing, and academic research contexts where evidentiary precision — not narrative assertion — is the standard.

3.2 Dataset Architecture

The data stack is organized into four layers. Each layer has a defined scope, unit of analysis, and governance rule. Layers do not substitute for one another.

Layer 1 — Sovereign Raw Evidence Archive

Repository: `bdi-raw-data-vault` (GitHub: IAMGODIAM) **Unit of analysis:** Series-level observations — annual, decennial, event-count, and aggregate records **Content:** Approximately 14,811 verified data points across 17 active files **Rule:** Primary source data is committed unmodified before any analysis. No scoring at this layer. All transformations happen downstream. **Sources include:** BLS Local Area Unemployment Statistics, U.S. Census Bureau (ACS, CPS, decennial), NCHS National Vital Statistics, BJS Prisoners series, Slave Voyages Database, NAEP Main Assessment, CFPB HMDA, Mapping Police Violence, EPA EJScreen (via Public Environmental Data Partners), USDA FARA

Layer 2 — Pillar Metric Layer (National Structural Record)

Repository: `bdi-sovereign-dataset` **Unit of analysis:** Flagship pillar metrics — disparity ratios, percentage-point gaps, absolute burden measures **Content:** 1,574 verified empirical observations across 8 pillars **Empirical window:** 1991–2024 (scoring series); 1514–1866 (Slave Voyages historical context — separate framing, separate pipeline) **Formula:** See Section 3.4

Layer 3 — Place-Level Compound Distress Layer

Repository (tract): `farmblock-data` — 15,507 census tracts, 49 cities **Repository (county):** `farmblock-dataset` — 24-county pilot published; ~3,144-county estimated full scope across 17 states **Unit of analysis:** FDI composite score (0–100) per tract or county **Formula:** See Section 3.4

Layer 4 — Forensic Narrative and Analytical Record

Repository: `bdi-raw-data-vault/reports/` **Content:** Analytical reports, claim triage records, scientific adjustment memos, peer review documentation

The architecture enforces a principle of evidentiary separation: raw data is never modified, transformation logic is documented in methodology files, and canonical numbers for public use are locked in the Stack Truth Table (see Section 3.4).

3.3 Data Sourcing

3.3.1 BDI National Structural Record — Eight Pillars

Pillar 1 — Economic - Federal Reserve Survey of Consumer Finances (SCF): wealth data, 1989–2022 vintages - U.S. Census Bureau Current Population Survey (CPS): poverty rates, 1991–2022 - U.S. Census Bureau American Community Survey (ACS): homeownership, income, 2022–2023 vintages - BLS Local Area Unemployment Statistics: annual unemployment by race, 1991–2024

Pillar 2 — Health - NCHS National Center for Health Statistics: life expectancy by race, 1991–2022; maternal mortality rates, 2018–2024 (Hoyert 2024) - CDC PLACES: small-area modeled estimates for chronic disease burden (diabetes, hypertension, obesity), 2023 vintage

Pillar 3 — Criminal Justice - BJS Prisoners series: incarceration rates by race, 1980–2022 - Mapping Police Violence: police killing rates by race, 2013–2024 - Sentencing Project: supplementary incarceration ratio documentation

Pillar 4 — Education - NAEP Main Assessment: 4th and 8th grade reading and math scores by race, 1992–2022

Pillar 5 — Housing - U.S. Census Bureau / CPS: homeownership by race, decennial and annual vintages - CFPB HMDA: mortgage denial rates by race, 2023 public dataset - Graetz et al. (2023) PNAS: eviction and eviction filing rates by race

Pillar 6 — Environmental - EPA EJScreen (via Public Environmental Data Partners, screening-tools.com/epa-ejscreen; note: removed from EPA.gov February 5, 2025) - Tessum et al. (2021) and Mikati et al. (2018): supplementary corroboration

Pillar 7 — Political and Civic - U.S. Census Bureau Current Population Survey November supplement: voter registration and turnout by race - NCSL and Brennan Center: voting rights restriction documentation

Pillar 8 — Historical Structural Context - Slave Voyages Database (slavevoyages.org, 2023 edition): 12.5 million Africans embarked; 10.7 million disembarked in the Americas; first recorded voyage 1514 - Federal Reserve / NCRC: redlining and lending discrimination studies - FHWA and academic literature: urban highway displacement documentation

3.3.2 FDI Place-Level Instrument — Tract Layer

SOURCE	VARIABLES	VINTAGE
Census Bureau ACS 5-Year	Poverty rate, median household income, internet access, housing vacancy rate, racial composition	2023
CDC PLACES	Diabetes prevalence, hypertension prevalence, obesity prevalence	2023
USDA FARA (proxy)	Food desert designation	2019 proxy via ACS income/poverty thresholds

Food desert proxy note: USDA FARA direct download was unavailable (HTTP 404 at time of collection). The proxy applied is: `food_desert_proxy = 1 WHERE poverty_rate ≥ 20% AND median_household_income ≤ $65,000`. This approximates the USDA Low-Income, Low-Access (LILA) standard. All rows are labeled to distinguish proxy-designated from directly confirmed designations.

3.3.3 FDI Place-Level Instrument — County Layer

SOURCE	VARIABLES	VINTAGE
BLS Local Area Unemployment Statistics	County unemployment rate	2025

SOURCE	VARIABLES	VINTAGE
Census Bureau ACS 5-Year	Poverty rate, income, racial composition	2022
CDC PLACES	Health burden indicators	2023
Census Bureau ACS 5-Year	Internet access, housing vacancy	2022

3.4 Validation Architecture — Stack Truth Table and Claim Triage Protocol

The Stack Truth Table

The project maintains a canonical crosswalk document — the Stack Truth Table (`STACK_TRUTH_TABLE.md`, committed to `bdi-raw-data-vault`) — that resolves conflicts between repositories, methodology files, READMEs, and paper language. The governing rule: when any document conflicts with the Stack Truth Table, the Stack Truth Table wins and the other document is updated.

The Stack Truth Table locks the following canonical public claims:

CLAIM	LOCKED VALUE	SOURCE
Tracts scored (FDI v2.0)	**15,507**	`processed/farmblock_fdi_v2.csv` row count, re-verified August 2026 (supersedes 15,578)
Cities ranked (FDI v2.0)	**49**	`processed/farmblock_city_rankings.csv` row count, re-verified August 2026 (supersedes 50)
Cities in county dataset	**53**	`processed/farmblock_cities_v2.csv` row count, verified
Counties in Phase 2 pilot	**24**	`farmblock_fdi_phase2.csv` row count, verified
Counties in Phase 3 estimated scope	**~3,144**	`farmblock_phase3_manifest.json` coverage field
BDI empirical observations	**1,574**	Verified JSON count per BDI_QUANT_SPEC.md counting rule
Raw vault observations	**~14,811**	VAULT_MANIFEST.json v3.1
BDI empirical window	**1991–2024**	BDI_QUANT_SPEC.md §2
Historical evidence anchor	**1514**	Slave Voyages DB, first recorded voyage
Priority states (BDI composite)	**17**	bdi_sovereign_dataset_v1.json state_scores

Data point counting rule (BDI_QUANT_SPEC.md §3): One data point equals one row in a time series or one unique observation record. JSON metadata keys (source, pillar, notes) are explicitly excluded. The earlier public claim of 1,855 data points included metadata keys as data points — this was incorrect. The corrected figure is 1,574 verified empirical observations.

The BDI Claim Triage Matrix

All public claims in research outputs are classified against four evidentiary categories before publication:

- **source-confirmed:** claim matches primary source at the stated vintage; safe to publish as-is with footnote
- **source-conflicted:** claim appears in draft but does not match primary source value, geography, or vintage; requires rewrite per triage guidance before publication
- **internally-derived:** claim is derived from E5 Enclave methodology not yet published externally; requires methodology publication or claim replacement before external sharing
- **citation-vintage-correction:** primary source is correct but URL or institutional location has changed (e.g., EPA EJScreen removed from EPA.gov, February 5, 2025; now via Public Environmental Data Partners)

The current Claim Triage Matrix (`BDI_CLAIM_TRIAGE_MATRIX.md`, committed to both `bdi-raw-data-vault` and `farmblock-data`) documents 27 claims across all eight pillars:

STATUS	COUNT
source-confirmed	10
source-conflicted	11
internally-derived	4
citation-vintage-correction	2
Total	**27**

Five highest-priority fixes required before external sharing:

1. **Slave Voyages (HI-1):** Distinguish "12.5 million embarked" from "10.7 million disembarked in the Americas" — these are different figures. Current draft conflates them.
2. **Cancer Alley geography (H-3/CD-5):** 85% Black is a fence-line census tract figure, not a parish-level figure. Parish level ranges from 22–57% Black. Specify geography to prevent factual challenge.
3. **NAEP gap (ED-1):** Current draft cites 20-point gap. Verified NAEP 2022 figure is 24 points. Parity timeline in draft is also incorrect by a factor of approximately two.

4. **Humphreys County poverty (CD-6):** Draft uses 37% (ACS 2019 vintage). Current figure is 32.1% (ACS 2022). Child poverty figure of 55% (ACS 2022) is more powerful and current. Recommend updating to current vintage.
5. **Section 3 HUD "$4.5 billion gap" (HO-3):** No published source exists for this aggregate. Replace with HUD OIG (2013) finding: 53% of PHAs did not file required Section 3 reports.

Dual-Source Verification Protocol

All primary claims in the BDI National Structural Record are verified against at least two independent sources before commitment to the vault. The `dual_source_log.json` file in `bdi-raw-data-vault` records the source pair, verification date, and any discrepancy notes for each verified series. Series that cannot be verified against two independent sources are flagged as single-source and labeled accordingly in the dataset.

FDI Quality Assurance

The FDI tract pipeline applies the following validation checks before publication: - Dimension values confirmed within 0–1 normalization bounds - **Sentinel-value audit (added August 2026):** eight tracts were found carrying imputation artifacts — `poverty_rate` filled to 100.0 alongside `median_hh_income` filled to the corpus median of $75,738.50. One of them, Census Tract 106.06 in Muscogee County, Georgia, is the highest-scoring tract in the published corpus (FDI 79.42) and is a group-quarters tract. These rows are enumerated in Appendix E and flagged for exclusion in v3.0. Because the instrument min–max normalizes across the corpus, an artificially maximal tract compresses every other tract's normalized position on the affected dimensions. - FDI composite confirmed between 0–100 - D4 health burden: confirmed using CDC PLACES direct measurement (diabetes + hypertension mean); not proxied in v2.0. Where CDC PLACES data is unavailable for a tract, D4 is imputed to zero — a conservative choice documented in LIMITATIONS.md. - Food desert proxy: all rows labeled (`food_desert_proxy_flag`) distinguishing proxy-assigned from FARA-confirmed designations

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3.5 Scoring Formulas

3.5.1 BDI National Composite (Product A)

`BDI_composite = Economic×0.20 + Health×0.20 + CriminalJustice×0.20 + Education×0.15 + Housing×0.10 + Environmental×0.10 + Political×0.05`

Weighting rationale: - Economic (20%), Health (20%), Criminal Justice (20%) carry equal highest weight — these three pillars document the most irreversible structural outcomes: wealth deprivation, mortality, and state-imposed incarceration - Education (15%) — a high-leverage pillar that compounds across generations - Housing (10%) — foundational wealth-building mechanism; weighted below the top three because homeownership data carries ACS survey variance at some geographies - Environmental (10%) — growing evidentiary base; weighted moderately given recent EPA data access disruption (February 2025) - Political (5%) — civic participation data carries the highest measurement noise of any pillar

All pillar scores normalized 0–100. Final composite normalized 0–100 across the comparison universe.

3.5.2 FDI Tract Composite (Product B — Tract Layer)

`FDI_tract = (D1_economic + D2_income_deficit + D3_food_access + D4_health + D5_vacancy + D6_digital) / 6 × 100`

Where: - D1: `poverty_rate` (Census ACS 2023) - D2: `1 − normalize(median_hh_income)` (inverted; lower income = higher distress) - D3: `food_desert_proxy × 0.60 + pct_no_internet × 0.40` - D4: `mean(pct_diabetes, pct_high_bp)` — CDC PLACES 2023 direct measurement - D5: `vacancy_rate` (Census ACS 2023) - D6: `pct_no_internet` (Census ACS 2023)

All dimensions min-max normalized 0–1 across all tracts in the dataset before compositing. Equal weighting (1/6 each) is the v2.0 baseline. PCA-derived weighting is recommended for v3.0.

3.5.3 FDI County Composite (Product B — County Layer, Phase 2)

`FDI_county = poverty×0.25 + health×0.25 + digital×0.20 + vacancy×0.15 + Black_pct×0.15`

Note on racial composition variable: The county-level formula includes `Black_pct` at 15% weight as a structural exposure proxy — not as a demographic descriptor or outcome variable. The theoretical basis is that communities with higher Black population concentration have historically been subjected to greater disinvestment, redlining, and structural exclusion. The variable captures exposure to systemic risk, not any intrinsic characteristic of race. This choice is documented in `methodology/race_variable_note.md`. The tract-level formula does not include this variable; the tract formula uses structural dimensions only.

3.6 Cross-Dataset Reconciliation

The two products share geography but operate at different resolutions. Reconciliation rules:

Temporal alignment: The BDI scoring series (1991–2024) and the FDI cross-sectional snapshots (ACS 2022–2023, BLS 2025) use different temporal frames by design. The BDI answers "what happened over time." The FDI answers "where is distress concentrated now." They are complementary, not substitutable.

Geographic alignment: FDI tract and county scores feed city and state aggregates used for cross-referencing against BDI state-level composite scores. Where city boundaries do not align with county or tract boundaries, the larger administrative unit is used and labeled.

Product A/B separation: The BDI national composite score and the FDI tract/county scores are distinct instruments. They share source data (CDC PLACES, ACS) but use different formulas, different geographic units, and different purposes. They must not be cited interchangeably. The Stack Truth Table "Do NOT cite for" column governs this boundary.

3.7 Methodological Choices — Explicit Rationale

Equal weighting (FDI tract): FDI v2.0 uses equal dimension weights as a transparent, auditable baseline. Equal weighting makes no assumption about which structural burden matters more than another — a defensible position for a v1 public release. PCA-derived or expert-assigned weighting is recommended for v3.0 but would require public disclosure of the weight-assignment process.

Analyst-assigned weighting (FDI county and BDI composite): The county FDI and BDI composite use analyst-assigned weights with documented rationale. These weights are disclosed in the methodology files and are subject to sensitivity testing. Alternative weight sets produce similar rank orderings in high-distress communities, which increases confidence in the core findings.

Proxy substitution (food desert): The USDA FARA data download was unavailable at collection time. The ACS-based proxy is documented, labeled, and subject to validation against USDA's published LILA criteria. Tracts designated as food deserts via proxy are distinguishable from any FARA-confirmed designation in the dataset. This is consistent with accepted practice in small-area data estimation.

Normalization approach: Min-max normalization within each dataset's geographic universe ensures scores are interpretable as relative distress rankings within the study population. It does not support cross-study comparisons without re-normalization. This is labeled explicitly on all published outputs.

Inclusion of racial composition in county formula: Discussed in §3.5.3. The variable captures structural exposure, not intrinsic racial characteristics. Full methodological rationale is published in `race_variable_note.md`.

3.8 Limitations

1. **USDA FARA vintage (2019):** Food access conditions may have changed post-2019. ACS proxy substituted and labeled at the row level.
2. **CDC PLACES modeled estimates:** CDC PLACES data represents small-area model estimates, not direct clinical measurement. These are the best available tract-level health burden data but carry model uncertainty.

3. **D4 health imputed to zero:** Where CDC PLACES data is unavailable for a tract, D4 is set to zero (conservative). This may understate distress in data-sparse tracts.
4. **Equal weights (FDI tract):** Not empirically validated. No dimension is demonstrated to matter more than another at the tract level. PCA-derived weighting is the recommended path for v3.0.
5. **Geographic coverage:** The FDI tract dataset covers 49 cities (15,507 tracts). It is not a national random sample. National coverage with weighting adjustments is planned for v3.0.
6. **ACS 5-year rolling window:** ACS 5-year estimates represent rolling averages, not point-in-time snapshots. Temporal precision is limited compared to administrative records.
7. **Correlation only:** Both the BDI and FDI identify co-occurrence of structural conditions. Neither instrument models causation. All analytical language uses co-occurrence framing.
8. **EPA EJScreen access disruption:** EJScreen was removed from EPA.gov on February 5, 2025. Current access is via Public Environmental Data Partners (screening-tools.com/epa-ejscreen). Environmental pillar data should be reverified using this updated source before final publication.
9. **Internally-derived compound distress scores:** Compound distress scores for flagship counties are derived from E5 Enclave methodology rather than from a federal instrument. *Status resolved for this edition:* the published Humphreys County score is **87.25** (an earlier draft figure of 83.5 was superseded), and the full component breakdown — formula, weights, normalization method, component scores, and the 7.5-point discrepancy between the published and hand-calculated values — is printed in Appendix E.

3.9 Data Governance and Ethics

License: All data products are released under CC0 1.0 Universal (Public Domain). No restriction on use, distribution, or modification. No attribution required, though attribution is appreciated.

Intended beneficiaries: The datasets are designed for use by community organizations, legal advocates, academic researchers, policymakers, journalists, and grant writers working in communities affected by the structural conditions the data documents. The data belongs to the communities it describes.

Black-first framing: Per E5 Enclave's LOGOS Doctrine, all language describing affected communities avoids deficit framing. Communities are not described primarily by what they lack. Structural critique names systems and their documented effects, not populations.

Evidentiary standard: The BDI Claim Triage Matrix (§3.4) enforces a pre-publication evidentiary review for all public claims. Conflicted claims are rewritten before external sharing. Internally-derived claims are either supported with published methodology or replaced with verifiable source-confirmed alternatives.

Institutional authority: E5 Enclave Incorporated, EIN 99-3822441, is the publishing entity. All decisions about canonical numbers, formula choices, and public language rest with the board. The Stack Truth Table is the authoritative governance document for all numerical claims.

3.10 Summary

This research produces two architecturally linked instruments built on a four-layer sovereign data stack. The BDI National Structural Record documents 1,574 verified empirical observations across eight pillars and 33 years. The FDI Place Distress Instrument scores 15,507 census tracts across 49 cities and 24 counties in a published pilot release.

Both instruments are correlation tools, not causal models. Both are released CC0 into the public domain. Both are governed by a Stack Truth Table and Claim Triage Matrix that enforce numerical consistency across all repositories and research outputs.

The methodology is designed to withstand peer review. The limitations are disclosed. The formula choices are documented with rationale. The canonical numbers are locked and governed.

The data is built to be used — by the communities, advocates, and researchers whose work these numbers serve.

3.11 Reconciliation Performed for This Edition (August 2026)

Three classes of check were run before this edition was set in type, and each produced corrections that are carried into the text.

Recomputation. Every ratio, gap, sum, mean and projection appearing in the drafts was recalculated from the Layer-1 raw series rather than copied forward from prior drafts. This surfaced ten errors that no source check would have caught, because each involved correct source data combined incorrectly. They include an unemployment "floor" claim contradicted by seventeen years of the series, a police-killings total understated by 481, a parity projection off by a factor of 3.3, and an incarceration floor stated 0.1 below the window minimum. All are enumerated in Appendix H.

Live source verification. Flagged series were re-pulled from primary federal sources in August 2026. The BLS unemployment series reproduced the vault values exactly. The NAEP Grade 8 reading series, queried against the NCES data service, confirmed the triage matrix's ED-1 ruling to the decimal: the 1992 gap was 29.63 points and the 2022 gap 24.54, against a draft claim of 20 points.

Governance-document audit. The Stack Truth Table's locked counts were tested against direct row counts of the published files. Two locked values — 15,578 tracts and 50 cities — were found stale, and the truth table had explicitly instructed that the pipeline manifest be corrected upward to match them. The row counts show the manifest was right and the table was wrong: the correct figures are 15,507 and 49, following the removal of duplicated Selma, Alabama rows at manifest version 2.1.

This last finding is the most instructive. The Stack Truth Table was created precisely to prevent numerical drift between repositories, and it became a source of drift because it was locked at a moment in time and the pipeline continued to improve beneath it. **A governance document that cannot be re-derived from the artifacts it governs will eventually contradict them.** Version 3.0 of this stack will generate the truth table from row counts at build time rather than maintaining it by hand.

CHAPTER 4

THE MEASURE: WHAT GETS COUNTED AND WHY

> "It is a peculiar sensation, this double-consciousness, this sense of always looking at one's self through the eyes of others."
>
> — W.E.B. Du Bois, *The Souls of Black Folk* (1903)

Opening

Measurement choices are moral choices. What you choose to count, what weight you assign each dimension, and what you decide to leave out of the composite is not a neutral technical exercise. It is an argument about what matters. Every index that has ever been built to measure human conditions — the United Nations Human Development Index, the Gini coefficient, the Federal Poverty Line — encodes a theory of what constitutes deprivation and what threshold marks its presence.

This chapter makes the measurement choices of the BDI explicit. It names the theory behind the weighting. It explains why eight dimensions were chosen and not three or twelve. It documents what the index does not claim to capture. And it argues that the specific combination of dimensions, weights, and data sources assembled here is not arbitrary — it is the minimum necessary architecture to see the wound whole.

I. Why Eight Dimensions

The decision to organize the BDI around eight structural pillars responds to a specific failure in how Black American structural distress has been measured historically: the single-pillar problem.

The federal government measures Black unemployment (BLS). It measures Black incarceration (BJS). It measures Black maternal mortality (NCHS). Each instrument is rigorous within its lane. None of them communicate with each other. A community can have low unemployment and high incarceration simultaneously. A community can have improving life expectancy and worsening mortgage denial rates simultaneously. The single-pillar instruments are not wrong. They are incomplete. They cannot see the interaction effects.

The academic literature on racial inequality has produced powerful multi-dimensional analyses — Robert Sampson's work on neighborhood effects, Patrick Sharkey's research on urban poverty, Keeanga-Yamahtta Taylor on predatory inclusion in housing markets. But this scholarship is paywalled, time-bounded by research funding cycles, and produced for academic audiences, not for the communities it describes.

The BDI synthesizes across eight dimensions simultaneously because the lived experience of structural distress is not single-pillar. A family in one of the highest-distress counties ranked in Chapter 7 is not experiencing poverty or health burden or carceral threat — they are experiencing all of these simultaneously, interacting with each other, compounding each other, foreclosing paths out of each other. An instrument that cannot see the compound cannot measure the catastrophe.

II. The Eight Pillars: Rationale and Evidence

Pillar 1 — Economic (Weight: 20%)

What it measures: Wealth gap, unemployment, household income, homeownership rates, poverty rates.

Why 20%: The economic dimension is assigned the highest weight (tied with health and criminal justice) because it is both the most measurable and the most foundational. Wealth is the mechanism through which structural advantage and disadvantage are transmitted across generations. The Federal Reserve Survey of Consumer Finances provides the most rigorous longitudinal measure available. The BLS unemployment series provides the longest continuous racial comparison available in American public data (1972–2025, fifty-four years). Income and homeownership from the Census ACS provide the intermediate layer between wealth accumulation and daily economic survival.

Key sources: Fed Reserve SCF (wealth, 1989–2022), BLS CPS (unemployment, 1972–2025), Census ACS B19013 (income), Census ACS B25003 (homeownership), Census ACS B17001 (poverty), BLS LAUS (state-level, 2010–2024), Census ACS county-level 5-year estimates (2015–2022, 12,382 county-year observations).

Pillar 2 — Health (Weight: 20%)

What it measures: Maternal mortality rate, life expectancy gap, COVID excess mortality, environmental health burden.

Why 20%: The body is the final ledger of everything the built environment, the economic environment, and the political environment produce. Maternal mortality is not a health failure — it is a system failure. The 69.9 per 100,000 maternal mortality rate for Black women in 2021 — the highest recorded since 1968, documented by NCHS — is a metric that encodes discriminatory clinical treatment, under-resourced prenatal care, environmental stressors, economic precarity, and the cumulative physical burden of navigating anti-Black institutions across a lifetime. Life expectancy encodes the same compound burden in a single number at the population level.

Key sources: NCHS WONDER (maternal mortality, 1999–2021), NCHS vital statistics (life expectancy by race, 1999–2021), CDC PLACES (health burden at tract level for FarmBlock layer).

Pillar 3 — Criminal Justice (Weight: 20%)

What it measures: Incarceration rate by race, police killing rate by race, unarmed killing rate.

Why 20%: The carceral system is an economic system. Incarceration removes labor from families, destroys wealth-building capacity, removes voting rights in most states, restricts housing access, restricts employment access, and generates lifetime economic penalties that compound across generations. The incarceration ratio — which has moved 0.14 points in ninety-seven years of data, and has not fallen below 5.8 within the empirical window — is not a public safety metric. It is a structural extraction metric. The BJS Prisoners series measures it with the same methodological rigor as any other federal statistical program.

Key sources: BJS Prisoners series (incarceration rate by race, 1991–2023), Mapping Police Violence database (police killings by race, 2013–2023, DOJ-confirmed methodology).

Pillar 4 — Education (Weight: 15%)

What it measures: NAEP reading and mathematics score gaps by race, educational attainment rates by race and state.

Why 15%: Education is weighted at 15% rather than 20% not because it matters less, but because the causal pathway from structural conditions to educational outcomes is more mediated than in the economic, health, and criminal justice pillars. School funding inequities, environmental stressors in learning environments, and the cumulative cognitive burden of poverty are all documented mechanisms. But the NAEP score gap is a downstream measure of upstream structural failures that are more directly captured in the higher-weighted pillars. The 15% weighting reflects this position in the causal chain, not a diminished importance.

Key sources: NAEP (reading and math score gaps by race, 1992–2022), Census ACS B15002 (educational attainment by race, all states, 2022).

Pillar 5 — Housing (Weight: 10%)

What it measures: Mortgage denial rates by race, historical homeownership rates, eviction rates.

Why 10%: Housing is the primary mechanism of wealth accumulation for American families without inherited wealth. The mortgage denial ratio — documented by the Home Mortgage Disclosure Act — is the most direct federal measure of discriminatory access to this mechanism. The denial ratio, which ranged from 2.01 to 2.36 across thirty years of HMDA reporting and never fell below 2.0, is one of the most durable findings in the dataset. The 10% weighting reflects the significant overlap between housing and economic conditions (Pillar 1): they are measuring different mechanisms of the same fundamental condition of wealth exclusion.

Key sources: HMDA (mortgage denial by race, 1995–2022), Census decennial homeownership (1940–2010 historical series), eviction data integrated from available administrative sources.

Pillar 6 — Environmental (Weight: 10%)

What it measures: Proximity to industrial pollution sources, environmental health burden by race and geography.

Why 10%: Environmental racism — the pattern by which polluting industrial facilities are sited disproportionately in Black and low-income communities — is documented across decades of environmental justice research and confirmed in federal EPA data. The Louisiana petrochemical corridor — 150 industrial facilities across an 85-mile stretch, with fence-line census tracts 65 to 94 percent Black against a parish-wide share near 44 percent — is the most documented example, but the pattern is national and appears in the FarmBlock data across the 50-city priority universe. The 10% weighting reflects that environmental burden is both a direct health input (it belongs to Pillar 2 in terms of outcomes) and a distinct structural condition in its own right (community geography as a site of intentional disinvestment and industrial siting).

Key sources: EPA facility data (industrial siting by community demographics), CDC PLACES environmental burden indicators at tract level.

Pillar 7 — Political (Weight: 5%)

What it measures: Elected representation by race at state and federal levels, voting access indicators.

Why 5%: Political representation is weighted at 5% — the lowest weight in the composite — not because political power is unimportant, but because the relationship between representation and structural outcomes is the most complex and least directly measurable in the BDI framework. The empirical record on the relationship between Black political representation and structural conditions in Black communities is contested in ways that the relationships in the other six pillars are not. A lower weight reflects appropriate epistemic humility about what the available data can support.

Key sources: Congressional and state legislative representation data by race, voting access indicators from existing civil rights monitoring databases.

Pillar 8 — Historical (Weight: 0% in composite, full presence in paper)

What it measures: The Trans-Atlantic slave trade, 1514–1866 — the origin point.

Why 0% composite weight: The historical pillar is not weighted in the BDI composite index because it is not a contemporary condition that varies across the 50-city comparison universe — it is the shared origin of the structural conditions that do. Every community in the priority universe descends from this history. Weighting it in the composite would not differentiate communities; it would add a constant. The historical data is present in the dataset, documented in the vault, and present in this paper because the 33-year empirical window does not explain itself without it. The wealth gap that opens Chapter 5 does not begin in 1991.

Key sources: Slave Voyages Trans-Atlantic Slave Trade Database (1514–1866), 42 aggregate observations: 12,521,337 embarked, 10,702,656 disembarked in the Americas, 1,818,681 lost in the crossing.

III. The BDI Composite Index Formula

The BDI composite score for a given geography (national, state, metro, or county) is calculated as:

```
BDI = (Economic × 0.20) + (Health × 0.20) + (CriminalJustice × 0.20) + (Education × 0.15)
+ (Housing × 0.10) + (Environmental × 0.10) + (Political × 0.05)
```

Each pillar score is normalized to a 0–100 scale using min-max normalization across the comparison universe before weighting. A BDI score of 100 represents the most extreme structural distress in the comparison universe; a score of 0 represents the least. The composite is not an absolute measure — it is a relative instrument for identifying and ranking the most structurally compromised geographies within the priority universe.

IV. What the BDI Does Not Measure

Mental health burden. The psychological toll of navigating anti-Black institutions — the documented phenomenon of weathering, the elevated cortisol levels and accelerated biological aging associated with chronic racial stress — is real, measurable, and consequential. It is not fully captured in the BDI v1.0 because the tract-level mental health data required to include it with the rigor this instrument demands was not available at time of synthesis. BDI v2.0 must address this.

Intergenerational wealth transfer. The Fed Reserve SCF captures current wealth levels. It does not fully capture the compound interest of historical exclusion — the way that the denial of land ownership to Black families during the Homestead Act era, the destruction of the Greenwood District in Tulsa, the targeted displacement of Overtown in Miami, and similar events represent not just historical injuries but compound wealth deficits that accumulate across generations. Quantifying this requires a different kind of modeling than the BDI framework employs.

Criminal legal debt. Court fees, fines, and supervision costs constitute a significant mechanism of wealth extraction from Black communities that is not captured in standard incarceration data. The BJS Prisoners series counts people. It does not count the dollar amounts extracted from communities through the criminal legal system. This is a gap.

Sub-county variation. The sovereign dataset covers national, state, MSA, and county levels. The FarmBlock layer addresses the census tract. The gap between county and tract — the neighborhood level, the school district level, the block level — is partially addressed but not fully resolved.

These gaps are documented because a sovereign empirical record that conceals its limitations is not sovereign — it is promotional. The findings of this paper stand on what the data does show. They are sufficient.

V. The Transition

Chapter 3 described how the data was built. Chapter 4 has described what it measures and why.

Part Three begins with the findings. Thirty-three years. The wealth gap that has never closed. The unemployment ratio that has never once inverted in fifty-four years. The maternal mortality ratio that is wider today than it was under legal segregation.

The data is ready. The architecture is described. The methodology is documented.

Turn the page.

PART THREE

The Findings

CHAPTER 5

THE ECONOMIC WOUND

> "The problem of the twentieth century is the problem of the color line."
>
> — W.E.B. Du Bois, *The Souls of Black Folk* (1903)

The wealth gap between Black and white American families has not closed. Not in 1991. Not in 2008. Not in 2024. Not in any year in between. This chapter documents that statement with four instruments — wealth, unemployment, income and homeownership — and one finding that runs through all four: the ratios move, sometimes substantially, while the absolute distance grows.

5.1 Wealth: The Ratio Nearly Tripled and the Gap Is the Widest on Record

The Federal Reserve's Survey of Consumer Finances is conducted every three years and is the most rigorous longitudinal measure of American household wealth in existence. Twelve waves span 1989 to 2022.

Table 5.1 — Median family wealth by race, Federal Reserve SCF, 1989–2022 *All figures in thousands of constant 2022 dollars. Source: Federal Reserve Board, FEDS Notes, "Greater Wealth, Greater Uncertainty: Changes in Racial Inequality in the Survey of Consumer Finances," October 18, 2023, Figure 2. Unit of observation: families.*

YEAR	BLACK MEDIAN	WHITE MEDIAN	ABSOLUTE GAP	RATIO
1989	$9.20k	$164.03k	$154.83k	0.056
1992	$20.51k	$144.42k	$123.91k	0.142
1995	$21.13k	$148.33k	$127.20k	0.143
1998	$28.26k	$174.75k	$146.49k	0.162
2001	$32.45k	$205.53k	$173.08k	0.158
2004	$32.09k	$221.47k	$189.38k	0.145
2007	$30.14k	$245.11k	$214.97k	0.123
2010	$21.80k	$178.28k	$156.48k	0.122
2013	$16.78k	$180.68k	$163.90k	0.093
2016	$21.09k	$210.94k	$189.85k	0.100
2019	$27.97k	$218.14k	$190.17k	0.128

YEAR	BLACK MEDIAN	WHITE MEDIAN	ABSOLUTE GAP	RATIO
2022	**$44.89k**	**$285.01k**	**$240.12k**	**0.158**

Three findings, and they do not point the same way.

Black median wealth grew faster than white median wealth, by a wide margin. In constant dollars it rose from $9,200 to $44,890 — a 388 percent increase. White median wealth rose from $164,030 to $285,010, or 74 percent. Black family wealth grew more than five times faster in percentage terms across the thirty-three years. The ratio moved from 5.6 cents on the dollar to 15.8 cents: it nearly tripled.

And the absolute gap is wider in 2022 than in any other year of the series. It grew from $154,830 to $240,120 — an increase of $85,290 in real terms, a 1.55-fold expansion. Not one of the eleven earlier waves records a larger real gap.

Both statements are true, from the same twelve rows. Faster percentage growth on a base one-eighteenth the size does not close a distance; it widens it. A family starting at $9,200 that quintuples its wealth ends at $44,890. A family starting at $164,030 that grows 74 percent ends at $285,010. The first family did far better proportionally and finished $85,000 further behind in absolute terms than it started.

That is the arithmetic of compounding inequality, and it is the central mechanism of this chapter. It recurs in every pillar that follows: the rate improves, sometimes dramatically, and the distance grows.

A correction to this table, and what it cost

The wealth table printed in earlier editions of this paper was wrong, and the error was material. It is corrected above and the correction is worth stating in full, because of how it was found.

Earlier editions gave 1989 as $12,000 Black and $95,000 white, with a gap of $83,000, and reported that the gap had "nearly tripled" to $240,100 — a 2.89-fold increase. Those 1989 figures were nominal, the table never said so, and the 2022 endpoint was the Federal Reserve's constant-2022-dollar figure. The series was a splice: a real endpoint back-filled with nominal history. Its apparent trend was substantially an artifact of thirty-three years of inflation.

The problem is not only the price basis. Deflating the old 1989 values to 2022 dollars yields $224,211 white and $28,321 Black — against the Federal Reserve's published $164,030 and $9,200. The old figures do not reconcile with the Fed's race-specific medians under *either* basis, by $60,181 and $19,121 respectively. They came from some other series or definition, and their provenance was never documented. The whole table has therefore been rebuilt directly from the Federal Reserve's published figures, which are cited above by title, date and figure number.

Note where the two series agree exactly: 2022. That agreement is the diagnostic. Whoever assembled the original table took the Fed's current real endpoint and extended it backward from an incompatible source.

This error survived the recomputation audit described in Chapter 3 and Appendix H, because that audit verified that every derived figure followed correctly from the vault's stored series. It did — the arithmetic was right. What it did not ask was whether the stored series was itself coherent. An internal consistency check cannot catch an incoherent input, and this paper's audit protocol has been amended accordingly: primary-source reconciliation of every series endpoint is now a required pass, not an optional one.

The correction was identified by an independent review of the printed edition, conducted with no access to this project's repositories. That is the process working as intended, and it is the reason this paper is released under a licence that permits exactly such review.

On projecting parity

Earlier editions stated that ratio parity was "approximately 868 years" away. That figure was computed from the flawed nominal series and is withdrawn. But it should not simply be replaced with a corrected single number, because the corrected number is not stable.

Extrapolating the observed rate of ratio change from 1989 to 2022 gives 274 years to parity. Extrapolating the same way from other baseline years in the same table gives:

BASELINE	IMPLIED YEARS TO PARITY
1989	274
1992	1,632
1995	1,511
2001	no convergence — the ratio is lower in 2022 than in 2001
2007	366
2013	117

The answer moves by a factor of fourteen depending on which year one starts from, and one plausible baseline yields no convergence at all. The 1989 wave is a particular problem: the ratio rises from 0.056 to 0.142 in the single step to 1992, a 2.5-fold move that no subsequent wave approaches, which makes 1989 a fragile anchor for any trend line.

No parity horizon is asserted in this paper. A linear extrapolation of two endpoints is an arithmetic operation on history, not a statement about the future, and when the operation is this sensitive to its inputs it does not support a headline figure. What the table does establish, without extrapolation, is that after thirty-three years the ratio stands at 15.8 cents on the dollar and the absolute gap has never been larger. Those are measurements. They are enough.

The land beneath the wealth

Wealth data begins in 1989 because that is when the SCF begins. The asset base it measures has a longer history.

Table 5.2 — Black-owned farmland, USDA Census of Agriculture, 1910–1997

YEAR	ACRES	CHANGE FROM PEAK
1910	15,000,000	peak
1940	11,000,000	−26.7%
1960	6,000,000	−60.0%
1978	3,100,000	−79.3%
1997	1,500,000	**−90.0%**

Thirteen and a half million acres — ninety percent of the peak — transferred out of Black ownership across eighty-seven years. No single confiscation event. Tax sales, heir-property partition actions, discriminatory USDA lending, foreclosure, harassment and violence, each documented, each individually small enough to escape public memory.

This is the largest systematic asset transfer in American history without a conventional name.

Two notes on this series. The Census of Agriculture changed methodology in 2017 to include leased acreage, producing an apparent recovery to 4.7 million acres and then 2.9 million in 2022; these are not comparable to the pre-2017 ownership series and are excluded from the trend statement above. Separately, the peak year is contested in the literature: 15 million acres is commonly dated to 1910, as here, but some compilations place the maximum nearer 1900 and others give a somewhat higher 1910 total. The 1997 figure of 1.5 million acres is well supported by that year's Census of Agriculture; the 90 percent loss should be read as measured against a peak whose exact year and level carry some uncertainty.

5.2 Unemployment: Fifty-Four Years, and the Ratio Has Never Inverted

The BLS series LNS14000006 and LNS14000003 — Black and white unemployment, seasonally adjusted, monthly — begin in 1972. Fifty-four years of continuous, race-disaggregated federal labor data. Nothing else in the American statistical system runs this long at this frequency.

Table 5.3 — Black/white unemployment ratio, BLS, selected years

YEAR	BLACK	WHITE	RATIO	
1972	10.40%	5.05%	2.059	series opens

YEAR	BLACK	WHITE	RATIO	
1983	19.50%	8.37%	2.330	
1989	11.47%	4.48%	**2.560**	maximum
2000	7.59%	3.48%	2.181	
2007	8.26%	4.12%	2.005	
2009	14.78%	8.49%	1.741	recession
2019	6.07%	3.27%	1.856	
2020	11.57%	7.34%	**1.576**	minimum
2022	6.11%	3.19%	1.915	
2024	—	—	1.666	
2025	6.90%	3.73%	1.850	

Fifty-four-year distribution: minimum 1.576 (2020), maximum 2.560 (1989), mean 2.115, median 2.130, standard deviation 0.230. The ratio equalled or exceeded 2.0 in 37 of 54 years.

Aggregation rule. Each annual ratio is computed from the twelve unrounded monthly seasonally-adjusted rates, not from the rounded annual averages displayed in the table. The two methods differ in the second decimal place: 2023's monthly averages are 5.5167 and 3.2750, giving 1.685, whereas dividing the displayed 5.5 by 3.3 gives 1.667. The unrounded method is used throughout, and readers reproducing these figures from published annual averages should expect small differences on that account. Figures are also subject to BLS vintage revision.

Two claims that appeared in earlier drafts do not survive this table, and both are withdrawn here.

The first was that Black unemployment "has never fallen below twice the white rate." It has — in seventeen of fifty-four years. The second, from an internal adjustment memo, was that the ratio never falls below 1.86 in a normal economic year. It does: 1.683 in 2023 and 1.666 in 2024, both ordinary expansion years, the narrowest sustained readings in the series.

What survives is stronger for being exact. Across fifty-four years, four recessions, one pandemic, nine presidencies and every configuration of American economic policy in the modern era, **the ratio has never inverted.** Not for one year. Not for one month. Black unemployment has never once been lower than white unemployment in the entire history of the federal government measuring both.

A ratio that oscillates between 1.58 and 2.56 and never crosses 1.0 is not noise around a fair baseline. It is a bounded quantity — and bounded quantities have mechanisms.

Note also the shape of the minimum. The lowest ratio on record, 1.576, occurs in 2020 — and it occurs because *white* unemployment surged to 7.34%, not because Black unemployment fell. The ratio narrowed in the worst labor market in eighty years. Convergence toward the mean is not always good news; sometimes it means everyone is drowning and the distance between swimmers has shrunk.

5.3 Income and Poverty

Table 5.4 — Median household income by race, Census ACS

YEAR	BLACK	WHITE	RATIO
2005	$31,869	$52,176	0.611
2010	$33,578	$54,168	0.620
2015	$36,898	$62,000	0.595
2019	$41,935	$68,785	0.612
2022	$48,200	$75,400	0.639

Seventeen years; 2.8 points of ratio improvement. The absolute income gap grew from $20,307 to $27,200.

Table 5.5 — Poverty rate by race, Census ACS

YEAR	BLACK	WHITE	RATIO
2005	25.6%	9.0%	2.84
2010	27.1%	10.6%	2.56
2015	25.4%	10.4%	2.44
2019	21.2%	9.0%	2.36
2022	21.3%	9.5%	2.24

The Black poverty rate fell 4.3 points across the series — real improvement, and it should be named as such. The ratio narrowed from 2.84 to 2.24 and did not approach 2.0 in any year of the ACS record. One in five Black Americans lived below the federal poverty line in 2022, in the wealthiest economy in human history.

5.4 Homeownership, and What the Fair Housing Act Did Not Do

Homeownership is where the previous three sections converge. It is the primary mechanism by which American families without inherited capital accumulate wealth, and it is the mechanism most directly governed by a specific federal civil rights statute.

Table 5.6 — Homeownership by race, Census decennial and ACS

YEAR	BLACK	WHITE	GAP (PP)	SOURCE
1940	23.6%	45.7%	22.1	Census of Housing
1950	34.5%	57.0%	22.5	Census of Housing
1960	38.4%	64.4%	26.0	Census of Housing
1970	41.6%	65.4%	23.8	Census of Housing
1980	44.4%	67.8%	23.4	Census of Housing
1990	43.4%	68.2%	24.8	Census of Housing
2000	46.3%	73.8%	27.5	Census of Housing
2010	44.3%	73.0%	28.7	Census of Housing
2022	44.1%	73.0%	**28.9**	ACS
2023	44.7%	72.4%	27.7	ACS

The Fair Housing Act was signed in April 1968. The gap that year was approximately twenty-four percentage points. Fifty-four years later it was 28.9.

The law did not close the gap. The gap is wider than when the law was passed. This is the single most important finding in the housing pillar, and it is not a claim about intent — it is a claim about outcome, which is the only thing a dataset can adjudicate.

An earlier draft placed the current gap at 30.1 points. The verified ACS 2022 figure is 28.9, and the 2023 release puts it at 27.7. The corrected figures are used throughout.

Access to the mechanism

Table 5.7 — Mortgage denial rate by race, HMDA

YEAR	BLACK	WHITE	RATIO
1993	34.3%	15.3%	2.24
2000	24.5%	10.4%	2.36
2003	20.3%	10.1%	**2.01**
2010	26.1%	12.4%	2.11
2019	17.4%	8.5%	2.05

YEAR	BLACK	WHITE	RATIO
2022	19.8%	9.1%	2.18

Thirty years of Home Mortgage Disclosure Act reporting. The denial ratio ranged from 2.01 to 2.36 and **did not fall below 2.0 in any year**. Denial rates for both groups fell substantially — 34.3% to 19.8% for Black applicants — while the ratio between them held almost perfectly flat.

2008

The subprime crisis is the clearest case in this dataset of a mechanism operating in plain sight.

Black homeownership peaked around 2004 and reached its trough in 2013 — a decline of roughly nine years, not the eighteen months an earlier draft asserted. The 18-month framing is withdrawn; the correct statement is that **the crisis reversed nearly a decade of accumulated homeownership gains across the 2007–2013 window**, and that Black median wealth fell to $11,200 in 2013, its lowest point in the entire SCF series.

The loans that produced it were not distributed randomly. They were marketed, and the marketing was documented at the time by the regulators who did not stop it.

Section 3

Section 3 of the Housing and Urban Development Act of 1968 requires that HUD-funded construction generate employment and contracting opportunities for low-income residents of the communities being developed. It is a statutory mandate, not a goal.

A 2013 audit by HUD's own Office of Inspector General found that **53 percent of public housing authorities did not file the required Section 3 reports** — which means that for more than half the program, whether the statutory benefit was ever delivered is not merely unproven but unknowable.

Earlier drafts of this paper cited a "$4.5 billion cumulative Section 3 gap." No published source supports that aggregate; it was internally derived and its construction was never published. It has been removed. The HUD OIG finding replaces it, and is stronger for coming from HUD.

5.5 What the Economic Pillar Establishes

Four instruments, one pattern. In every case the rate improves and the distance grows. In every case the ratio between Black and white outcomes proves more stable than any policy intervention aimed at it. In no case does a legal remedy — the Fair Housing Act, HMDA disclosure, the Section 3 mandate — produce convergence in the measure it was written to address.

The economy is not failing to deliver equity. It is delivering, reliably, across fifty-four years, a distribution it was built to deliver.

Chapter 6 turns to the body, the cell block and the classroom, where the same architecture produces outcomes that cannot be recovered.

CHAPTER 6

THE HEALTH WOUND, THE JUSTICE WOUND, THE EDUCATION WOUND

> "Not everything that is faced can be changed, but nothing can be changed until it is faced."
> — James Baldwin

Chapter 5 measured what is extracted. This chapter measures what cannot be given back: years of life, years of liberty, years of instruction. These are the irreversible pillars. A wealth gap can in principle be closed by transfer. A woman who dies in childbirth is not recoverable by any policy.

PART I — THE BODY

6.1 The Maternal Mortality Reversal

This is the single most disturbing series in the dataset, and it must be read in order.

Table 6.1 — Maternal mortality per 100,000 live births, NCHS, 1915–2022

YEAR	BLACK	WHITE	RATIO
1915	1,050.0	600.0	1.75
1930	**900.0**	**609.0**	**1.48**
1940	773.5	319.8	2.42
1950	221.6	61.1	3.63
1960	97.9	22.4	4.37
1970	59.8	14.4	4.15
1980	21.5	6.7	3.21
1990	22.4	5.1	4.39
2000	22.6	7.5	3.01
2010	28.4	12.7	2.24
2019	44.0	17.9	2.46
2021	**69.9**	26.6	2.63
2022	49.5	19.0	**2.61**

American medicine made childbirth dramatically safer. In 1930, roughly nine women in a thousand died giving birth if they were Black; by 2022 the figure was one in two thousand. That is a genuine and enormous achievement of the twentieth century.

Now read the ratio column.

In 1930 — under legal Jim Crow, in segregated hospitals, with the explicit denial of equivalent care — a Black woman died in childbirth 1.48 times as often as a white woman. In 2022, she died 2.61 times as often.

The ratio did not narrow across ninety-two years of modern medicine. It widened by a factor of 1.76.

The mechanism is differential improvement. Medicine got better for everyone and better for white women faster, and it did so continuously enough that the disparity ratio is now higher than it was under a healthcare system that did not pretend to serve Black women equally.

A necessary caveat on the 1930 comparator. Before 1933 the United States had no complete national birth registration area; the figures for 1915 and 1930 derive from the registration states only, and the racial composition of those states was not representative of the country. The 1930 ratio of 1.48 is therefore a weaker measurement than the 2022 ratio of 2.61, and the two are not strictly commensurable. The 2022 figures are confirmed directly against NCHS: 49.5 per 100,000 for Black women and 19.0 for white women, a ratio of 2.605, conventionally rounded to 2.61.

The cross-era comparison is offered because the direction is robust across every intermediate decade in the table — the ratio is higher in 2022 than in 1930, 1940, 1980, 2000 and 2010 alike — not because the 1930 datum is precise. Readers who reject the pre-1933 figures entirely still face a ratio that rose from 2.24 in 2010 to 2.61 in 2022, within a fully registered national system.

The 2021 peak — **69.9 deaths per 100,000** — is the highest Black maternal mortality rate recorded in the United States since 1968. The World Health Organization associates rates in that range with low-income countries. It occurred in the wealthiest nation in human history, in the twenty-first century, three years ago.

The 2022 figure of 49.5 represents real improvement from the pandemic peak, and NCHS reporting indicates continued decline thereafter. Improvement without erasure: the rate falls, the ratio holds.

6.2 Life Expectancy, and What COVID Revealed

Table 6.2 — Life expectancy at birth by race, NCHS, 1900–2022

YEAR	BLACK	WHITE	GAP (YEARS)
1900	33.0	47.6	14.6

YEAR	BLACK	WHITE	GAP (YEARS)
1940	53.1	64.2	11.1
1960	63.6	70.6	7.0
1980	68.1	74.4	6.3
2000	71.8	77.3	5.5
2010	74.7	78.8	4.1
2019	74.8	78.8	**4.0**
2020	71.8	77.6	5.8
2021	70.8	76.4	5.6
2022	72.8	76.5	3.7

Across 122 years the gap narrowed from 14.6 years to 3.7 — the most substantial convergence anywhere in this dataset, and it should be stated plainly before anything else is said about it.

Then read 2010 to 2021. Nine years of effort moved the gap by one-tenth of one year: 4.1 to 4.0. Two years of pandemic moved it by 1.8 years in the opposite direction.

Black life expectancy fell from 74.8 years in 2019 to 70.8 in 2021 — four full years, returning the figure to a level last seen in the mid-1990s.

Earlier drafts described this as "twenty years of gains erased in eighteen months." The direction is right and the arithmetic was not: the decline occurred across two calendar years, and it returned Black life expectancy to roughly its mid-1990s level — approximately two decades of accumulated gains, lost over two years. That is the corrected formulation and it is used throughout.

A pandemic is a stress test. It does not create the structure it exposes; it loads it until the weakest joins fail first. The 2022 recovery to a 3.7-year gap — the narrowest on record — indicates the system can move when it is forced to. It moved 1.8 years in the wrong direction in twelve months and 1.9 in the right direction in twelve more. Nine years of ordinary policy moved it one-tenth of a year.

6.3 The Fence Line

Environmental burden is where health outcomes acquire coordinates.

The Louisiana petrochemical corridor along the Mississippi between Baton Rouge and New Orleans — commonly called Cancer Alley — contains roughly 150 industrial facilities across an 85-mile stretch. Cancer incidence and respiratory illness in the corridor exceed state and national averages.

A geographic precision is required here, and earlier drafts did not supply it. The claim that the corridor is "85 percent Black" is not accurate at the parish level, where Black population share ranges from about 22 to 57 percent — St. James Parish, the most-studied case, is approximately 44 percent Black.

The 85-percent figure describes something narrower and more damning: the **fence-line census tracts**, the communities immediately adjacent to the facilities themselves. At that resolution, in St. James Parish Districts 4 and 5, **Black population share ranges from 65 to 94 percent** (ACS 2022).

The parish-level number does not capture where the burden lands. The tract-level number does. The distinction matters because it is exactly the distinction industrial siting decisions operate on — and because a reviewer who checks the parish figure and finds 44 percent will discard the entire chapter. The corrected geography is stated everywhere this claim appears.

Source note: EPA's EJScreen tool was removed from EPA.gov on February 5, 2025. It remains publicly accessible via the Public Environmental Data Partners archive, which is the citation used here. The removal of a federal environmental justice screening tool during the period this paper documents is itself a data point about the durability of public records — and part of why this dataset is sealed on-chain.

PART II — THE CELL

6.4 Ninety-Seven Years of a Constant

Table 6.3 — Imprisonment rate per 100,000 by race, BJS, 1925–2022

YEAR	BLACK	WHITE	RATIO
1925	142	22	6.45
1940	272	50	**5.44**
1960	661	96	6.89
1980	1,111	168	6.61
1990	2,376	339	7.01
2000	3,457	449	**7.70**
2010	3,074	459	6.70
2014	2,724	466	5.85
2018	2,272	392	5.80
2020	1,882	296	6.36
2022	1,862	295	6.31

In 1925, Black Americans were imprisoned at 6.45 times the white rate. In 2022, at 6.31 times.

The ratio moved 0.14 points in ninety-seven years.

Between those two readings: two world wars, the Great Migration, the New Deal, *Brown v. Board*, the Civil Rights Act, the Voting Rights Act, Thurgood Marshall on the Supreme Court, the entire NAACP legal campaign, the War on Drugs, mass incarceration's rise and partial retreat, the First Step Act, and *The New Jim Crow* becoming required reading in every progressive institution in the country.

The absolute rates tell a second story. Black imprisonment rose from 142 to a peak of 3,457 per 100,000 and has since fallen to 1,862 — a genuine and substantial decarceration since 2000, roughly 46 percent off the peak. White imprisonment followed nearly the same arc. The decline is real; the ratio between the two is what refused to move.

Two corrections, and one of them is unresolved.

Earlier drafts stated the ratio "has never fallen below 5.7." Within this table the minimum is **5.80, in 2018**, and the full-series range is **5.44 to 7.70**. Those are the corrected bounds.

The second correction is more serious and is disclosed here rather than settled. **The series above does not match the Bureau of Justice Statistics' published all-adults imprisonment rates.** BJS reports, for 2022, an imprisonment rate of **1,196 per 100,000 Black adult U.S. residents and 229 per 100,000 white adult residents — a ratio of 5.22**, not 6.31. The table's 1,862 and 295 sit far above those figures and correspond closely to BJS's *male* imprisonment rates (approximately 1,826 and 279 per 100,000 same-race male residents).

The likely explanation is that this series measures male imprisonment rates, or uses a total-population rather than adult-population denominator. The vault does not record which, and the pre-1980 figures come from a separate historical compilation (Cahalan 1986) whose crosswalk to the modern BJS series is not documented. Until that crosswalk is published, the correct statement of scope is narrower than earlier editions claimed:

> *On the all-adults basis that BJS publishes, Black adults were imprisoned at **5.22 times** the white rate in 2022. On the basis used in the series above*
>
> — which appears to be male rates — the 2022 ratio is 6.31. Both exceed five. The paper does not assert which basis its historical series uses, because that has not been established.

The finding that survives either reading is the **stability**: a ratio between roughly five and eight for as long as the United States has measured it, moving 0.14 points across the ninety-seven years of the compiled series. That stability is what the chapter argues from, and it does not depend on the denominator. But a paper that prints "Black Americans" where its source may say "Black males" is overstating its scope, and this edition stops doing so.

Required for the next edition: the exact BJS table identifiers, numerator and denominator definitions, and sex and age restrictions for every endpoint in this series, plus the Cahalan-to-BJS crosswalk.

When a system produces the same ratio across a century of legal, political and cultural transformation, that ratio is not a byproduct. It is an output specification.

6.5 Police Killings

Table 6.4 — Police killings by race, Mapping Police Violence, 2013–2023

YEAR	BLACK KILLED	TOTAL KILLED	BLACK SHARE	UNARMED BLACK
2013	312	1,108	28.2%	35
2015	346	1,307	26.5%	57
2018	282	1,165	24.2%	27
2020	312	1,145	27.2%	28
2021	328	1,134	28.9%	29
2023	337	1,286	26.2%	26
Total 2013–2023	**3,496**	**13,096**	**26.7%**	**397**

Three thousand four hundred ninety-six Black Americans killed by police in eleven years. Three hundred ninety-seven of them unarmed.

Earlier drafts reported 3,015 and 377. Those figures were undercounts; the totals above are the summed annual rows from the source database and are the corrected figures of record.

Black Americans were 13.6 percent of the population in the 2020 Census and 26.7 percent of those killed by police across the eleven-year window — an overrepresentation factor of approximately 1.96. The annual Black share ranged from 24.2 to 28.9 percent and shows no trend across a decade that contained Ferguson, the 2020 protests, dozens of consent decrees and the most sustained public attention to police violence in American history.

Instrument note: Mapping Police Violence is not a federal statistical program. The United States does not maintain a complete federal count of civilians killed by police. MPV is the most comprehensive public database in existence and is labeled as non-federal wherever cited in this paper. That the most complete record of lethal state force in America is maintained by volunteers is itself a finding.

PART III — THE CLASSROOM

6.6 What NAEP Shows

The National Assessment of Educational Progress is the most methodologically rigorous measure of American academic achievement, administered by the Department of Education on a common scale since 1990.

Table 6.5 — NAEP score gaps by race, national

ASSESSMENT	1992 GAP	2022 GAP	CHANGE
Grade 8 reading	29.6 pts	**24.5 pts**	−5.1
Grade 8 mathematics	40 pts	32 pts	−8
Grade 4 reading	32 pts	28 pts	−4

Grade 8 reading, verified against the live NCES data service in August 2026: in 1992 white students averaged 267.0 and Black students 237.4, a gap of 29.6 points. In 2022 the figures were 268.4 and 243.9 — a gap of **24.5 points**.

Two measurement notes. NCES discloses that the 1992 administration did not permit testing accommodations while later administrations did, so the endpoints are not perfectly comparable instruments. And the figures above carry more decimal places than NCES publishes in its rounded digest tables (267/237 and 268/244); the precision comes from the underlying data service, not from the printed tables.

Two corrections, both material. Earlier drafts stated the current gap at 20 points and projected parity in 64 years. The verified gap is approximately 24 points, and a mechanical linear extension of the observed rate — 5.1 points across thirty years, or 0.17 points annually — puts the remaining 24.5 points **roughly 144 years** out.

Not sixty-four. A century and a half — and, as with the wealth projection in Chapter 5, that figure is an arithmetic extension of two endpoints, not a forecast. It is stated to give the observed rate a human scale, not to predict a year. The children entering kindergarten this September would need to be succeeded by four more generations before the trend line, extended, reaches parity.

Both errors understated the severity, and both were trivially checkable by any education researcher. They are corrected here because a paper that overstates progress in the direction of comfort is making the same error as one that overstates distress in the direction of alarm.

The gap is closing. It is not closing at a rate that constitutes a solution, and the honest statement of that is more useful than either the optimistic error or the pessimistic one.

Note also the 2022 anomaly: Grade 4 reading widened from 26 points in 2019 to 28, and Grade 8 mathematics from 30 to 32. Pandemic learning disruption was not evenly distributed either.

6.7 What These Three Pillars Establish

Health, justice and education measure three different institutions with three different funding structures, three different governing statutes and three different professional cultures.

They produce the same shape. A ratio that proves more durable than the reforms aimed at it. Absolute improvement that is real and that does not close the distance. And in the maternal mortality series, the clearest case in the dataset of a disparity that is *worse* under formal equality than it was under formal segregation.

Chapter 7 stops measuring pillars separately and asks what happens in the places where all of them fail at once.

CHAPTER 7

THE COMPOUND CATASTROPHE

> "They have their plans set decades and scores in advance... 50 to 60 years ago these plans were on the books and it's coming to pass every day."
>
> — Ralph McCartney, Overtown, Miami, 1997

Eight pillars measured separately produce eight findings. Measured together, they produce something the separate measurements cannot see: places where every pillar has failed simultaneously, and where the failures feed each other faster than any single-pillar remedy can reach them.

A county with high poverty is a poor county. A county with high poverty, the highest hypertension prevalence in its comparison universe, no acute-care hospital, twenty-eight percent of its housing stock vacant and thirty percent of its residents without home internet is not a poor county. It is a county in which each condition forecloses the escape route from the others. No hospital means telehealth; no internet means no telehealth. Vacancy means no property tax base; no tax base means no hospital.

This chapter measures that interaction. It is also the chapter in which this paper's own instrument is subjected to the same scrutiny the instrument applies to everyone else.

7.1 The FarmBlock Distress Index

The FDI is a place-level composite operating at two resolutions.

County layer (24-county published pilot):

`FDI_county = poverty(0.25) + health_burden(0.25) + digital_exclusion(0.20) + vacancy(0.15) + Black_pct_proxy(0.15)`

Tract layer (15,507 tracts), which deliberately excludes racial composition entirely:

`FDI_tract = (D1_poverty + D2_income_deficit + D3_food_access + D4_health + D5_vacancy + D6_digital) / 6 × 100`

All dimensions are min–max normalized across the corpus before compositing. The two instruments are built differently on purpose: the county formula includes Black population share at 15 percent as a structural-exposure proxy; the tract formula uses structural dimensions only. Where the two agree, the agreement is not an artifact of the race variable, because one of them does not contain it.

The rationale for the county proxy is set out in full in Appendix E, along with the strongest objection to it — that including racial composition in an instrument used to demonstrate racial disparity risks circularity. The tract instrument exists partly as the answer to that objection.

Reproducibility status, stated before the scores rather than after them. The FDI outputs in this chapter are **not independently reproducible from the published package.** Public: the scored files, the formulas above, the weights, the normalization method, the Humphreys decomposition and the enumerated defects. Not yet public: the raw pre-scoring inputs with their ACS table identifiers and vintages, the full Phase 3 corpus that supplies the normalization bounds, and the transformation code with a dependency lockfile. A reader can therefore audit the *logic* of everything that follows and cannot yet audit its *arithmetic*. Every score in this chapter — 87.25 for Humphreys County, the 15,507 tract scores, the 49 city rankings — should be cited as this organization's published output, not as an independently verified result. This is the finding of the September 2026 independent review, accepted without qualification; the full statement is Appendix E.4a, and closing the gap is the first deliverable of FDI v3.0.

The chapter is retained on that footing rather than withdrawn, because a disclosed limitation a reader can navigate is worth more than a silence. What follows is offered as a published instrument with its audit trail attached and its weakest joint named — including, in §7.3, this project's own failed attempt to reproduce its flagship score.

7.2 The Ranked Counties

Table 7.1 — FarmBlock County Distress Index, top 12 of 24 published

RANK	COUNTY	FDI	% BLACK	POVERTY	MEDIAN INCOME	NO INTERNET	VACANCY	HYPERTENSION
1	**Humphreys, MS**	**87.25**	80.0%	35.0%	$24,000	30.0%	28.0%	51.7%
2	**Claiborne, MS**	**85.26**	83.0%	35.0%	$26,000	28.0%	25.0%	50.0%
3	**Sunflower, MS**	**78.34**	72.0%	32.0%	$28,000	26.0%	24.0%	48.0%
4	**Alexander, IL**	**77.38**	33.2%	21.4%	$40,365	38.2%	42.4%	47.9%
5	**Amite, MS**	**70.86**	40.2%	27.1%	$34,866	30.2%	19.3%	43.2%
6	East Carroll, LA	70.02	67.0%	40.3%	$30,856	29.7%	28.3%	49.6%
7	Adams, MS	68.31	53.4%	27.2%	$37,271	17.3%	22.0%	51.7%
8	Barbour, AL	65.68	46.9%	24.2%	$39,712	24.3%	22.8%	46.4%
9	Ashley, AR	59.55	24.8%	23.3%	$44,804	23.6%	23.7%	41.4%
10	Gadsden, FL	57.98	55.0%	22.0%	$45,000	18.0%	14.0%	40.0%

RANK	COUNTY	FDI	% BLACK	POVERTY	MEDIAN INCOME	NO INTERNET	VACANCY	HYPERTENSION
11	Wayne (Detroit), MI	56.79	40.0%	22.0%	$54,000	16.0%	20.0%	38.0%
12	St. Louis City, MO	55.43	44.0%	22.0%	$51,020	14.0%	18.0%	43.0%

A correction to this paper's own framing. Earlier drafts named five "compound catastrophe zones": Humphreys County, Detroit, East St. Louis, Claiborne County, and the Louisiana petrochemical corridor. That list does not match the instrument's output and is withdrawn.

The instrument's actual top five are **Humphreys, Claiborne, Sunflower, Alexander and Amite** — four Mississippi Delta counties and one Illinois river county. Wayne County (Detroit) ranks eleventh at 56.79, thirty points below Humphreys. East St. Louis sits in St. Clair County, Illinois, which is not in the published corpus at all; Alexander County, Illinois — rank four — is Cairo, a different place with a different history.

The five originally named were selected for documentary depth, not by score. Detroit and Cancer Alley belong in this paper, and they appear below as case studies with their actual standing disclosed. But a composite index that names a top five must name the top five the composite produced. Substituting the more familiar cities would have been the exact failure this instrument exists to prevent.

7.3 Humphreys County, Mississippi

FIPS 28053. Population approximately 7,400 and falling. FDI 87.25 — the highest score in the corpus.

Table 7.2 — Humphreys County score decomposition

COMPONENT	RAW VALUE	NORMALIZED	WEIGHT	CONTRIBUTION
Poverty	35.0%	1.000	0.25	0.250
Health burden	36.15% (mean of diabetes 20.6, hypertension 51.7)	1.000	0.25	0.250
Digital exclusion	30.0% no internet	0.447	0.20	0.089
Vacancy	28.0%	0.419	0.15	0.063
Structural exposure (% Black)	80.0%	0.964	0.15	0.145

Humphreys sits at the corpus ceiling on both of the highest-weighted components. **51.7 percent hypertension prevalence** — in a county where high blood pressure is the statistical majority condition, not the exception. The nearest acute-care hospital option disappeared when Humphreys County Medical Center closed in **2013**; thirteen years later, three in ten residents have no home internet, which means the telehealth substitute is unavailable to the same households.

The county qualifies as low-income, low-access under USDA Food Access Research Atlas criteria. *An earlier draft stated "no grocery store within fifteen miles." USDA FARA does not use a fifteen-mile threshold — its measures are 0.5, 1, 10 and 20 miles — and the claim is replaced with the designation the federal instrument actually issues.*

On poverty, three figures and why all three appear. ACS 2015–2019 recorded 36.4 percent all-persons poverty; ACS 2022 recorded 32.1 percent, with **55.0 percent of children** below the line; the most recent ACS five-year release reports approximately 27 percent, with a margin of error of ±6.3 points. Earlier drafts used "37 percent" without a vintage lock. All three vintages are printed here because in a county of 7,400 the survey margin is wide enough that a single figure asserted without its interval is not a finding — it is a rounding decision presented as one. The decline is also not straightforwardly good news: Humphreys has lost roughly a third of its population since 1980, and a poverty rate calculated on a shrinking denominator measures departure as much as improvement.

On the score itself. The published FDI is 87.25. Recomputing the five components by hand against approximate corpus bounds yields **79.7** — a gap of roughly 7.5 points, arising because the published score normalizes against the full Phase 3 corpus while a hand calculation uses approximate range endpoints. The published figure stands as the citation of record. The discrepancy is printed rather than smoothed because Humphreys is this paper's flagship example, and a flagship example that cannot be audited is a slogan.

7.4 The Cases the Index Ranks Lower

Detroit — Wayne County, Michigan (rank 11, FDI 56.79)

Wayne County's composite is thirty points below Humphreys, and the reason is instructive rather than exculpatory. Detroit has infrastructure: hospitals, broadband, a tax base, transit. Its distress is concentrated and internal rather than county-wide. The tract layer shows this directly — across **587 Detroit tracts** the median FDI is 33.9 while the maximum reaches **68.8**. The county average conceals a distribution in which specific neighborhoods score in Delta territory while others do not.

This is the strongest argument for the tract instrument. County-level analysis of a large metropolitan county will systematically understate concentrated urban distress, because the affluent tracts and the abandoned tracts are averaged into a single unremarkable number. Detroit's problem is not that the county is uniformly distressed. It is that the distressed parts are as distressed as rural Mississippi and are politically invisible inside a county average.

The Louisiana corridor

The petrochemical corridor does not appear as a single ranked unit because it is not a county — it spans several parishes along an 85-mile stretch, and the burden is concentrated in fence-line tracts rather than distributed across parish geography. East Carroll Parish, Louisiana ranks sixth at 70.02 with the highest poverty rate in the corpus (40.3 percent), but East Carroll is in the northeast Delta, not the industrial corridor.

The corridor's documentation is therefore tract-level and is presented in Chapter 6: fence-line census tracts in St. James Parish Districts 4 and 5 are 65 to 94 percent Black, against a parish-wide figure of approximately 44 percent. The instrument that captures this is the tract instrument, and the geography that matters is the tract geography.

7.5 The Tract Layer: 15,507 Tracts, 49 Cities

Table 7.3 — Highest-distress cities by median tract FDI

CITY	TRACTS	MEDIAN FDI	MAX FDI	MEDIAN POVERTY	MEDIAN % BLACK
Albany, GA	29	49.5	70.8	27.5%	70.6%
Jackson, MS	67	45.3	62.5	24.2%	85.7%
Macon, GA	48	41.9	74.1	26.8%	51.5%
Shreveport, LA	73	39.9	71.0	19.0%	52.4%
Bronx, NY	348	39.0	58.2	25.0%	32.0%
St. Louis, MO	104	38.6	67.2	21.0%	44.3%
Augusta, GA	56	37.6	62.0	19.7%	60.7%
New Orleans, LA	182	35.3	66.2	20.9%	58.5%
Detroit, MI	587	33.9	68.8	19.3%	31.0%
Baltimore, MD	199	33.4	65.8	19.7%	71.7%

The gap between median and maximum in every row is the finding. Macon's median tract scores 41.9 and its worst scores 74.1. Detroit's median is 33.9 and its worst is 68.8. **Distress is not distributed across these cities. It is concentrated inside them**, which is why place-based intervention has to be targeted below the city level to reach it.

7.5.1 Corrections to the published counts

Three canonical figures in this paper's own governance documents were found to be stale and are corrected here.

CLAIM	PREVIOUSLY LOCKED	VERIFIED
Tracts scored	15,578	**15,507**
Cities	50	**49** (48 distinct names; Columbus appears in both GA and OH)
Counties, tract layer	—	49

The Stack Truth Table of April 18, 2026 locked 15,578 tracts and 50 cities as "verified row counts," and explicitly instructed that the pipeline manifest's lower city figure be corrected upward to match. Direct row counts of the published files show the opposite: the manifest was right and the truth table was wrong. The manifest records the reason — **Selma, Alabama was removed at version 2.1 because its rows duplicated Montgomery County** — a deduplication performed after the truth table was written and never propagated to it.

Every downstream figure in this paper has been rebuilt on 15,507 and 49.

7.5.2 A defect in the published tract file

Eight of the 15,507 tracts carry an imputation artifact that inflates their scores, and one of them is the highest-scoring tract in the corpus.

All eight share a signature: `poverty_rate` set to exactly 100.0 alongside `median_hh_income` set to exactly $75,738.50 — the corpus median, a fill value. Several also carry `pct_no_internet` at 5.879292, likewise a corpus mean. These are tracts where the Census returned suppressed or null values, and the pipeline substituted a maximum for one field and a central-tendency fill for another.

The consequence is concrete. **Census Tract 106.06, Muscogee County, Georgia — the top-ranked tract in the entire dataset at FDI 79.42 — is one of these eight.** It records 100 percent poverty, 100 percent without internet, and a $75,738 median household income simultaneously, which is not a possible combination. It is a group-quarters tract on a military installation. Its inflated score also sets the reported maximum for Columbus, Georgia in the published city rankings.

Because the tract instrument min–max normalizes across the corpus, an artificially maximal tract compresses every other tract's normalized position on the affected dimensions. The eight rows are enumerated in Appendix E and flagged for exclusion in FDI v3.0.

A second, opposite-direction issue: **398 tracts — 2.6 percent of the corpus — have the D4 health dimension imputed to zero** where CDC PLACES data was unavailable. That choice is conservative and it understates distress in exactly the data-sparse places most likely to be distressed.

Neither defect changes the county rankings or any finding in Chapters 5 or 6. Both are disclosed because the alternative — publishing an index whose top-ranked unit is an artifact, and waiting to see whether anyone checks — is the practice this paper was built to refuse.

7.6 What Targeting Means

The FarmBlock instrument exists to answer an operational question: given finite capacity, where does intervention go first?

The answer the data gives is not the answer political attention gives. The highest-distress places in this corpus are small Delta counties with declining populations and no organized constituency — Humphreys, Claiborne, Sunflower, Amite — and one Illinois river county, Alexander, whose profile is driven by 42.4 percent housing vacancy and 38.2 percent digital exclusion rather than by poverty alone.

These places are not where cameras go. They are where the composite says the compound burden is heaviest.

That is the entire argument for building the instrument. A ranked, auditable, publicly available composite makes it possible to direct resources by measured burden rather than by visibility — and to be held accountable when the direction chosen does not match what the measurement says.

Chapter 8 turns to what the full record demands.

PART FOUR

The Argument

CHAPTER 8

WHAT THE DATA DEMANDS

> "Strong people don't need strong leaders."
> — Ella Baker

This is not a call for sympathy. It is a demand for accounting. The preceding chapters settled a factual question; this one addresses what follows from the answer.

8.1 The Political Pillar, and the Inflection of 2013

One series has been held back until now, because it is the pillar that governs whether any of the others can be addressed.

Table 8.1 — Voter turnout by race, Census CPS November supplement

YEAR	BLACK	WHITE	GAP (PP)
1964	58.5%	70.7%	−12.2
1976	48.7%	60.9%	−12.2
1984	55.8%	61.4%	−5.6
1996	50.6%	60.7%	−10.1
2004	60.0%	65.8%	−5.8
2008	64.7%	64.4%	**+0.3**
2012	**66.6%**	**64.1%**	**+2.5**
2016	59.6%	65.3%	−5.7
2020	62.6%	70.9%	−8.3

In 2012, for the first and only time in fifty-six years of recorded federal data, **Black voter turnout exceeded white voter turnout** — 66.6 percent against 64.1. It was the highest civic participation rate Black Americans have ever achieved in a measured American election.

In June 2013, the Supreme Court decided *Shelby County v. Holder*, removing the coverage formula in Section 4(b) of the Voting Rights Act and with it the preclearance requirement that had governed jurisdictions with documented histories of voting discrimination since 1965.

By 2016 the gap was −5.7 points. By 2020 it was −8.3 — wider than in 2004, before the Obama-era surge.

The data does not establish causation. Turnout responds to candidates, contests, mobilization budgets and a dozen other things; a single decision cannot be isolated as the cause of a multi-election decline, and this paper does not claim it. What the data establishes is sequence: the peak was 2012, the decision was 2013, and the two subsequent presidential elections produced the widest gaps since the 1990s. Whether that sequence is causal is a legal and social-scientific question. That it is the sequence is a matter of record.

This is why the political pillar carries only 5 percent weight in the composite — its measurement noise is the highest of any pillar, and epistemic humility about what turnout data can support is the honest position. It is also why it is discussed last and at length: it is the pillar that determines whether the other seven can be legislated about at all.

8.2 The Historical Architecture

Three figures anchor the record that precedes the empirical window. Each is stated at full precision, because precision is the argument.

12,521,337. Africans forced onto slave ships between 1514 and 1866, across roughly 36,000 documented voyages. Not "about twelve and a half million" — a counted figure, from a database of individual voyage records maintained across four decades of archival work.

1,818,681. Those who did not survive the crossing. A 14.5 percent mortality rate. They appear in no census of the enslaved because they never arrived. They exist in the record only as the difference between two numbers.

10,702,656. Those who reached the Americas. Of these, approximately **388,747** disembarked on the North American mainland — 3.6 percent of the total. That population reached roughly four million by 1860, a factor of about 10.3 across two centuries, through natural increase under conditions of legal chattel slavery.

Earlier drafts of this paper stated that "12.5 million Africans were transported to the Americas." That sentence conflates the embarked and disembarked figures and erases 1.8 million deaths. It is corrected everywhere it appeared.

To these three, a fourth from Chapter 5: **13,500,000 acres** of Black-owned farmland lost between 1910 and 1997 — ninety percent of the peak.

The wealth gap measured at $240,100 in 2022 is the present-tense reading of an instrument calibrated by these figures. No economic model in existence contains them.

8.3 What the Eight Pillars Demand

The findings do not translate into a single policy. They translate into a standard against which policies can be tested, and the standard has four parts.

Enforcement is the variable, not legislation

The pattern in Chapter 5 is not an absence of law. The Fair Housing Act (1968), the Home Mortgage Disclosure Act (1975), Section 3 of the HUD Act (1968) and the Voting Rights Act (1965) are all on the books. The homeownership gap is wider than when the Fair Housing Act passed. The mortgage denial ratio has not dropped below 2.0 in thirty years of mandatory disclosure. Fifty-three percent of public housing authorities did not file the Section 3 reports that would show whether the statute was being honored.

The measurable variable is not whether a remedy exists but whether anyone is required to demonstrate compliance with it. Any intervention proposed on the basis of this data should be evaluated first on its enforcement architecture and reporting requirement, because the historical record shows that a mandate without a filing obligation produces an unmeasurable and therefore unaccountable program.

Ratios are the metric, not rates

Every pillar in this paper shows absolute improvement. Poverty fell. Denial rates fell. Imprisonment fell 46 percent from its peak. Maternal deaths fell from 900 per 100,000 to 49.5. Life expectancy rose 39 years.

And in every pillar the ratio held, or widened. A program evaluated on absolute improvement will report success in every one of these domains while the disparity it was created to address is unchanged or worse. **The maternal mortality series is the decisive case: absolute performance improved by a factor of eighteen while the disparity ratio grew by a factor of 1.76.**

Any intervention derived from this data must be measured on the gap, not on the level. This is a methodological demand, and it is the one most often refused, because level metrics are easier to achieve and easier to report.

The unit of intervention is below the county

Chapter 7 established that county-level analysis systematically understates concentrated urban distress. Detroit's county composite is unremarkable; its worst tracts score in Delta territory. A funding formula that allocates by county will route resources away from the neighborhoods with the highest measured burden, because the county average is diluted by the tracts that do not need it.

The tract instrument exists for this reason, and its limitation is disclosed: it covers 15,507 tracts in 49 cities, not the nation. National tract coverage is the principal deliverable of version 2.0.

Interaction effects require simultaneous intervention

Humphreys County has no hospital and thirty percent of households without internet. Addressing either alone yields little: telehealth without connectivity is unavailable, and connectivity without a care endpoint is a browser. The compound structure means sequential single-pillar programs — a broadband grant this year, a clinic study next year — can each succeed on their own metrics while the compound burden does not move.

This is the operational case for place-based, multi-dimensional intervention, and it is the design premise of the FarmBlock program.

8.4 The Sovereign Standard

A word about why this record was built the way it was.

CC0, not open access. Open access means free to read under a license. CC0 means public domain: no license, no attribution requirement, no permission, no restriction. The organizer in Humphreys County does not need to cite anyone, ask anyone, or credit anyone to use this in a grant application.

On-chain, not merely archived. The dataset is sealed on Base Mainnet. This is not a technological flourish. During the period this paper documents, EPA's EJScreen environmental justice tool was removed from EPA.gov (February 5, 2025). Federal datasets documenting racial disparity have been contested, defunded and withdrawn. A record that exists only on servers controlled by institutions subject to political pressure is a record that can be edited by political pressure. This one cannot.

Unfunded, deliberately. Every condition of funding is a potential condition of finding. This dataset has no funder to satisfy and therefore one obligation: accuracy. That is also why Appendix H exists.

Corrected in public. Twenty-seven claims were triaged; eleven conflicted with their sources. Ten further arithmetic errors were found in the drafts by recomputation. Three canonical counts in the project's own governance documents were stale. The highest-scoring tract in the published index is an imputation artifact. All of it is printed.

A record that has never been corrected is a record no one has audited. The corrections are the credential, and any organization publishing a sovereign dataset should expect to publish its errata alongside it.

8.5 The Next Instrument

Version 2.0 requires four things, stated as commitments rather than aspirations:

1. **National tract coverage** — all eight pillars at census-tract resolution, beyond the current 49-city universe.
2. **The eight tracts excluded and the imputation logic rebuilt** — sentinel values must fail loudly rather than fill silently.
3. **PCA-derived or expert-assigned dimension weights**, with the assignment process published, replacing the equal-weight v2.0 baseline.
4. **Mental health burden and criminal legal debt** added as measured dimensions. Both are documented mechanisms of extraction that the current instrument cannot see.

8.6 What Is Being Asked

The data has settled whether the wound is real. It cannot settle what should be done, and this paper does not pretend that a spreadsheet can adjudicate a moral question.

What it can do is remove a specific excuse. It is no longer possible to say that the disparities are unmeasured, or that the measurement is behind a paywall, or that the record is fragmented across agencies that do not speak to one another, or that no one has assembled it, or that the community most affected has not documented its own condition.

It is measured. It is free. It is in one place. It is auditable, and its errors are printed with it.

What remains is a decision, and the decision does not belong to the data.

CONCLUSION

WHAT THE DATA DEMANDS OF WHOEVER HOLDS THIS PAPER

Thirty-three years in the core window, and series reaching back to 1900, to 1925, to 1940, to 1514. Eight dimensions of American life. Seventeen priority states. Forty-nine cities. Fifteen thousand five hundred and seven census tracts. Approximately fourteen thousand eight hundred and eleven raw federal observations, synthesized into one thousand five hundred and seventy-four verified empirical records, sealed on a public blockchain and released into the public domain.

The question before every person who holds this paper is no longer whether the wound is real. The data settled that. The question is what you intend to do now that you know.

What the Data Established

On wealth. In constant 2022 dollars, Black median family wealth grew 388 percent between 1989 and 2022 while white median wealth grew 74 percent — Black wealth rising more than five times faster in percentage terms. And the absolute gap widened, from $154,830 to $240,120, an increase of $85,290. The gap is larger in 2022 than in any other wave of the survey. Faster proportional growth on a base one-eighteenth the size does not close a distance; it widens it. That arithmetic is the mechanism this paper documents eight times over.

On labor. Across fifty-four years of monthly federal data — the longest continuous race-disaggregated labor series in American statistics — the Black/white unemployment ratio ranged from 1.58 to 2.56 and averaged 2.115. It has never once inverted. Not in one year of fifty-four, through four recessions, one pandemic and nine presidencies.

On housing. The Fair Housing Act was signed in 1968, when the homeownership gap stood at approximately twenty-four percentage points. In 2022 it was 28.9. The gap is wider than when the law was passed. Mortgage denial ratios have not fallen below 2.0 in thirty years of mandatory federal disclosure. And fifty-three percent of public housing authorities did not file the reports that would show whether a 1968 statutory mandate was ever honored.

On health. In 2021, Black women in America died in childbirth at 69.9 per 100,000 live births — a rate the World Health Organization associates with low-income nations. And the ratio that matters most in this dataset: in 1930, under legal Jim Crow and in segregated hospitals, a Black woman died in childbirth 1.48 times as often as a white woman. In 2022, she died 2.61 times as often. Ninety-two years of modern medicine widened the disparity by a factor of 1.76. Between 2019 and 2021, Black life expectancy fell four full years, to a level last seen in the mid-1990s.

On criminal justice. In 1925, Black Americans were imprisoned at 6.45 times the white rate. In 2022, at 6.31 times. The ratio moved 0.14 points in ninety-seven years, across two world wars, the Civil Rights Act, the Voting Rights Act, mass incarceration's rise and a genuine 46 percent decarceration from its peak. From 2013 to 2023, 3,496 Black Americans were killed by police; 397 were unarmed.

On education. The Grade 8 reading gap narrowed from 29.6 points in 1992 to 24.5 in 2022 — five points in thirty years. At the observed convergence rate, closing what remains requires approximately 144 years.

On political participation. In 2012, Black voter turnout exceeded white turnout for the first and only time in fifty-six years of federal record. In 2013, *Shelby County v. Holder* removed the Voting Rights Act's preclearance coverage formula. By 2020, the gap was −8.3 points, the widest since the 1990s. The sequence is documented. The causation is a legal argument this paper does not make.

On the historical record. Between 1514 and 1866, 12,521,337 Africans were forced onto slave ships. 1,818,681 did not survive the crossing. 10,702,656 reached the Americas. And between 1910 and 1997, thirteen and a half million acres of Black-owned farmland — ninety percent of the peak — passed out of Black hands without a single event large enough to be remembered by name.

What the Pattern Reveals

Across every pillar, in every decade, the same sequence.

A crisis is documented. Public pressure builds. A law is passed or a program funded. Enforcement infrastructure is underfunded and deprioritized. The metric improves in absolute terms — sometimes dramatically — while the ratio between Black and white outcomes holds or widens. The absolute improvement is reported as progress. Political energy dissipates. The structure continues.

This is not policy failure in the ordinary sense. Failure implies the intended outcome was not achieved. But the maternal mortality series makes the sharper reading unavoidable: absolute performance improved by a factor of eighteen while the disparity ratio grew by a factor of 1.76. A system can improve enormously and distribute the improvement unequally, and if it does so consistently for ninety-two years, the distribution is not an accident of the improvement. It is a property of the system doing the improving.

This is a claim about function, not intention. Systems are judged by what they produce. What this one has produced, measured across fifty-four years of labor data, ninety-seven years of incarceration data and one hundred and twenty-two years of mortality data, is a set of ratios more durable than every legal remedy aimed at them.

Ralph McCartney understood this from inside the geography. "*They have their plans set decades and scores in advance.*" He was not speaking metaphorically. The redlining maps confirm it. The urban renewal records confirm it. This paper confirms it at national scale across eight dimensions.

What This Paper Is

A forensic instrument, and a corrected one.

It documents what the system has done, where, and with what consistency. It maps the places where the compound weight of every pillar lands simultaneously — Humphreys, Claiborne, Sunflower, Amite, Alexander — and it names them by what the instrument measured rather than by what would have been more familiar to name.

It also documents its own errors. Twenty-seven claims triaged, eleven found conflicting with their sources. Ten arithmetic errors found by recomputing every derived statistic from the raw series. Three canonical counts in this project's own governance documents found stale. The highest-scoring tract in the published index found to be an imputation artifact. An unsourced $4.5 billion aggregate removed entirely. A "five compound catastrophe zones" list that did not match the instrument's output, withdrawn and replaced.

All of it is in Appendix H, with the original claim and the verified value side by side.

We print it because the alternative is a record that invites scrutiny it has not performed on itself. Every number in this paper can be checked against a public federal source or a public repository. Inviting that and concealing our own findings would be incoherent.

It is licensed CC0. It is sealed on-chain. It cannot be revised under pressure, placed behind a paywall, or defunded. It will be the same data in 2030 and in 2040, in the collections of institutions that do not yet exist.

The Call

Every generation of this tradition has faced the same question in a different form: knowing what we know, with the tools we have, in the moment we are in — what is the move?

The data answers the diagnostic question. It does not answer the strategic one. That answer lives in the communities this paper documents — in the organizing rooms, the churches, the HBCUs, the school board meetings, the courtrooms, the legislative chambers, and on the front porches of Humphreys County and Cairo and the Delta and Overtown and Liberty City.

McCartney was asked, near the end of his 1997 testimony, what he would want Overtown to be. He said: "*Where every child belongs to them and every child realizes that he has a parent in every adult. That's the type of community — sure it's ideal, but everybody looks for a haven.*"

He was describing what was taken. The data in this paper is the empirical record of the taking.

The wound is measured. The address is known. The lineage is long and the cost has been staggering. What the next chapter says depends on what is done with this instrument — by whom, in which rooms, toward which ends, and with what urgency.

The board submits this paper as evidence of what is, as testimony to what was, and as instrument for what must come.

By Grace, perfect ways.

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APPENDICES

APPENDIX A — SOURCE CITATION TABLE

Every series used in this paper, with the agency, instrument, coverage and vault location. All raw files were committed unmodified to `bdi-raw-data-vault` before any analysis was performed.

#	SERIES	AGENCY & INSTRUMENT	YEARS	VAULT FILE
1	Unemployment by race	Bureau of Labor Statistics CPS, series LNS14000006 / LNS14000003	1972–2025	`economic/bls_unemployment_by_race_1972-2025_FULL_RAW.json`
2	State unemployment	Bureau of Labor Statistics LAUS	2010–2024	`economic/tier1_state_bls_unemployment_2010-2024_RAW.json`
3	Homeownership, income	Census Bureau ACS 1-year, B25003B/H, B19013B/H	2005–2022	`economic/census_acs_homeownership_income_RAW.json`
4	Poverty	Census Bureau ACS 1-year, B17001B/H	2005–2022	`economic/census_acs_poverty_RAW.json`
5	Family wealth	Federal Reserve Board Survey of Consumer Finances, triennial	1989–2022	`economic/fed_reserve_scf_wealth_usda_land_RAW.json`
6	Black farmland	USDA NASS Census of Agriculture	1910–2022	`economic/fed_reserve_scf_wealth_usda_land_RAW.json`
7	State economics	Census Bureau ACS 5-year, all states	2010–2022	`economic/tier1_state_economics_ACS_2010-2022_RAW.json`
8	Metro economics	Census Bureau ACS 5-year, 516+ MSAs	2015–2022	`economic/tier2_metro_msa_economics_ACS_2015-2022_RAW.json`
9	County economics	Census Bureau ACS 5-year, 3,222 counties	2015–2022	`economic/tier3_county_economics_ACS5Y_2015-2022_RAW.json`
10	Life expectancy, maternal mortality	NCHS National Vital Statistics Reports	1900–2022	`health/nchs_life_expectancy_maternal_mortality_RAW.json`
11	Imprisonment by race	Bureau of Justice Statistics Prisoners series; Cahalan 1986 for pre-1980	1925–2022	`criminal_justice/bjs_incarceration_mpv_killings_RAW.json`
12	Police killings	Mapping Police Violence *(non-federal)* MPV public database	2013–2023	`criminal_justice/bjs_incarceration_mpv_killings_RAW.json`
13	NAEP score gaps	NCES Main Assessment, reading and mathematics	1992–2022	`education/tier1_naep_score_gaps_national_1992-2022_RAW.json`
14	Educational attainment	Census Bureau ACS B15002, all states	2022	`education/tier1_state_education_attainment_2022_RAW.json`
15	Historical homeownership; voter turnout	Census Bureau Census of Housing; CPS P20 November supplement	1940–2010; 1964–2020	`housing/census_decennial_homeownership_1940-2010_RAW.json`

#	SERIES	AGENCY & INSTRUMENT	YEARS	VAULT FILE
16	Mortgage denial; eviction	CFPB / FFIEC; Princeton Eviction Lab HMDA; Eviction Lab national dataset	1993–2022; 2000–2016	`housing/hmda_mortgage_denial_eviction_RAW.json`
17	Transatlantic slave trade	Slave Voyages, Emory University Trans-Atlantic Slave Trade Database, 2023 edition	1514–1866	`historical/slavevoyages_1514-1866_aggregate_RAW.json`
18	County Black population share	Census Bureau ACS B02001	2022	`demographics/tier3_county_black_population_pct_2022_RAW.json`

Supplementary sources cited in text: HUD Office of Inspector General, *Audit of HUD's Section 3 Program*, 2013-AT-0003 (March 28, 2013); Graetz et al., *PNAS* (2023), eviction and filing rates by race; EPA EJScreen via Public Environmental Data Partners (removed from EPA.gov February 5, 2025); Tessum et al. (2021); Mikati et al. (2018); CDC PLACES 2023; USDA Food Access Research Atlas 2019; HRSA Area Health Resources File.

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APPENDIX B — COUNTING METHODOLOGY

The rule. One data point equals one row in a time series, or one unique observation record.

Included: each year of an annual series; each census decade in a decennial series; each year × race combination in a stratified series; each year × grade combination in NAEP.

Excluded: JSON metadata keys (`source`, `pillar`, `notes`, `verification_note`); text-only records carrying no numeric observation; duplicate series committed from the same source in more than one file.

The correction of record. The dataset was publicly described at one point as containing **1,855 data points**. That count included metadata keys as though they were observations. Applying the rule above yields **1,574 verified empirical observations**. The corrected figure is used in every public statement, is recorded in the dataset JSON itself (`total_data_points_original_claim: 1855`, with a correction note), and appears in this paper's Introduction rather than only in a methods note.

Tier breakdown of the raw vault (~14,811 observations):

TIER	SCOPE	OBSERVATIONS
Tier 1	National series	479
Tier 1	State series	358
Tier 2	Metro / MSA (516+ areas)	1,550

TIER	SCOPE	OBSERVATIONS
Tier 3	County (3,222 counties)	12,382
Historical	1514–1866 aggregates	42

APPENDIX C — PILLAR SUMMARY

PILLAR	WEIGHT	HEADLINE FINDING	SERIES SPAN
Economic	20%	Real wealth gap widened \$154,830 → \$240,120 (widest on record) while the ratio nearly tripled, 0.056 → 0.158	1972–2025
Health	20%	Maternal mortality ratio 1.48 (1930) → 2.61 (2022); Black life expectancy −4.0 years, 2019–2021	1900–2022
Criminal justice	20%	Imprisonment ratio 6.45 (1925) → 6.31 (2022); moved 0.14 points in 97 years	1925–2023
Education	15%	Grade 8 reading gap 29.6 → 24.5 points; ~144 years to parity at observed rate	1992–2022
Housing	10%	Homeownership gap 28.9 pp in 2022 vs ~24 pp when the Fair Housing Act passed	1940–2023
Environmental	10%	Fence-line tracts, St. James Parish Districts 4–5: 65–94% Black vs ~44% parish-wide	2015–2022
Political	5%	2012 the only year Black turnout exceeded white (+2.5); −8.3 by 2020	1964–2020
Historical	context only	12,521,337 embarked; 1,818,681 lost; 13.5M acres of farmland lost 1910–1997	1514–1997

APPENDIX D — THE RANKED COMPOUND DISTRESS COUNTIES

FarmBlock County Distress Index, Phase 2 published pilot, n = 24.

RANK	COUNTY	FDI	RANK	COUNTY	FDI
1	Humphreys, MS	87.25	13	Acadia, LA	55.28
2	Claiborne, MS	85.26	14	Accomack, VA	54.37
3	Sunflower, MS	78.34	15	Benton, TN	51.59
4	Alexander, IL	77.38	16	Adams, WI	50.89
5	Amite, MS	70.86	17	Bronx, NY	50.43
6	East Carroll, LA	70.02	18	Adams, OH	48.85
7	Adams, MS	68.31	19	Tallapoosa, AL	48.36

RANK	COUNTY	FDI	RANK	COUNTY	FDI
8	Barbour, AL	65.68	20	Ashtabula, OH	45.70
9	Ashley, AR	59.55	21	Allegany, MD	43.14
10	Gadsden, FL	57.98	22	Essex (Newark), NJ	42.83
11	Wayne (Detroit), MI	56.79	23	Fresno, CA	40.03
12	St. Louis City, MO	55.43	24	Allen, IN	34.47

Earlier drafts named a "five compound catastrophe zones" list — Humphreys, Detroit, East St. Louis, Claiborne and Cancer Alley — that did not correspond to this ranking. Detroit ranks 11th. East St. Louis lies in St. Clair County, Illinois, which is not in this corpus. The list is withdrawn; the ranking above stands.

APPENDIX E — FARMBLOCK METHODOLOGY AND KNOWN DEFECTS

E.1 Formulas

County (Phase 2, n = 24): `FDI_county = normalize(poverty_rate) × 0.25 + normalize(mean(diabetes_pct, hypertension_pct)) × 0.25 + normalize(pct_no_internet) × 0.20 + normalize(vacancy_rate) × 0.15 + normalize(pct_black) × 0.15`

Tract (v2.0, n = 15,507): `FDI_tract = (D1_poverty + D2_income_deficit + D3_food_access + D4_health + D5_vacancy + D6_digital) / 6 × 100` where `normalize(x) = (x − corpus_min) / (corpus_max − corpus_min)`, and the tract formula contains **no racial composition variable**.

E.2 The `% Black` variable, and the objection to it

The county formula includes Black population share at 15 percent weight as a structural-exposure proxy: the theoretical basis is that communities with higher Black population concentration have been subjected to greater documented disinvestment, redlining and exclusion, and that the variable therefore captures accumulated exposure to systemic risk rather than any intrinsic characteristic.

The strongest objection is that this is partly circular. An index that includes racial composition and is then used to demonstrate racial disparity in outcomes has assumed part of what it sets out to show. This objection is legitimate and is not dismissed here.

Two responses, offered as arguments rather than refutations. First, the tract instrument excludes the variable entirely and produces convergent results — the highest-distress tracts are overwhelmingly majority-Black without the formula being told to look for that. Second, the county weights were sensitivity-tested and alternative weight sets produce similar rank orderings among high-distress counties. Readers who find the objection dispositive should use the tract instrument, which was built partly for them.

E.3 Humphreys County decomposition

COMPONENT	RAW	NORMALIZED	WEIGHT	CONTRIBUTION
Poverty	35.0%	1.000	0.25	0.250
Health burden	36.15%	1.000	0.25	0.250
Digital exclusion	30.0%	0.447	0.20	0.089
Vacancy	28.0%	0.419	0.15	0.063
Structural exposure	80.0%	0.964	0.15	0.145
Hand-calculated total				**0.797 → 79.7**
Published score				**87.25**

The published score normalizes against the full Phase 3 corpus; the hand calculation above uses approximate range endpoints, producing a 7.5-point difference. The published figure is the citation of record. The discrepancy is printed rather than reconciled away.

E.4 Known defects in the published tract file

Eight tracts carry imputation artifacts. Each has `poverty_rate` filled to exactly 100.0 alongside `median_hh_income` filled to exactly $75,738.50 — the corpus median. Several also carry `pct_no_internet` at the corpus mean. These are rows where the Census returned suppressed or null values and the pipeline substituted a maximum for one field and a central-tendency fill for another.

FIPS TRACT	CITY	POP	FDI	FILLS
13215010606	Columbus, GA	1815	79.42	poverty=100.0; income=corpus median; no_internet=100.0
36005031900	Bronx, NY	762	58.19	poverty=100.0; income=corpus median; no_internet=100.0
06037980014	Los Angeles, CA	40	58.08	poverty=100.0; income=corpus median
45079010806	Columbia, SC	876	45.41	poverty=100.0; income=corpus median; no_internet=corpus mean
45079010408	Columbia, SC	4243	43.01	poverty=100.0; income=corpus median; no_internet=corpus mean
04013981000	Phoenix, AZ	696	42.22	poverty=100.0; income=corpus median; no_internet=corpus mean
12057010900	Lakeland, FL	7218	31.37	poverty=100.0; income=corpus median

FIPS TRACT	CITY	POP	FDI	FILLS
12011980000	Jacksonville, FL	2	31.12	poverty=100.0; income=corpus median; no_internet=corpus mean

The first row is the highest-scoring tract in the entire published corpus. It reports 100 percent poverty, 100 percent without home internet, and a $75,738 median household income simultaneously — an impossible combination. It is a group-quarters tract on a military installation, and its score also sets the reported maximum for Columbus, Georgia in the published city rankings.

Because the tract instrument min–max normalizes across the corpus, an artificially maximal tract compresses every other tract's normalized position on the affected dimensions. These eight rows are flagged for exclusion in FDI v3.0, and the imputation logic will be rebuilt to fail loudly rather than fill silently.

398 tracts (2.6 percent) have D4 health imputed to zero where CDC PLACES data was unavailable. This is conservative in the wrong direction: it understates distress in precisely the data-sparse places most likely to be distressed.

Food-desert proxy. USDA FARA direct download returned HTTP 404 at collection time. The substitute is `food_desert_proxy = 1 WHERE poverty_rate ≥ 20% AND median_household_income ≤ $65,000`, approximating the USDA low-income/low-access standard. Every row carries a `food_desert_source` flag distinguishing proxy-assigned from FARA-confirmed designations.

E.4a Reproducibility status

The FDI outputs in this paper are not independently reproducible from the published package. That is the finding of the September 2026 independent review and it is accepted here without qualification.

What exists publicly: the scored output CSVs, the formulas, the weights, the normalization method, the Humphreys decomposition and the enumerated defects. What does not yet exist publicly: the raw pre-scoring inputs with ACS table identifiers, release years and geography codes; the full Phase 3 corpus that supplies the normalization bounds; the transformation code with a dependency lockfile; and a deterministic command that rebuilds every published score from raw inputs.

Until that package is released, a reader can audit the *logic* of the FDI and cannot audit its *arithmetic*. The distinction matters and it is stated plainly: the 87.25 score for Humphreys County, the 15,507 tract scores and the 49 city rankings should be cited as this organization's published outputs, not as independently verified results. Producing that package is the first deliverable of FDI v3.0.

E.5 Verified corpus counts

QUANTITY	VALUE	BASIS
Tracts scored	**15,507**	direct row count, `processed/farmblock_fdi_v2.csv`
Cities	**49**	direct row count, `processed/farmblock_city_rankings.csv` (48 distinct names; Columbus appears in GA and OH)
Counties, tract layer	49	pipeline manifest v2.1
Counties, published county pilot	24	direct row count, `farmblock_fdi_phase2.csv`

These supersede the Stack Truth Table values of 15,578 and 50, which were locked April 18, 2026 and did not incorporate the version 2.1 removal of duplicated Selma, Alabama rows.

APPENDIX F — PROVENANCE AND SEALS

ASSET	CONTRACT	TOKEN	NETWORK
BDI Sovereign Dataset v1.0	ExodusV4 `0x8582684C53912D496Df60C5B1B9Bb44D3d2f9B44`	#2	Base Mainnet

Repositories, all CC0 1.0 Universal:

REPO	LAYER	CONTENTS
`IAMGODIAM/bdi-raw-data-vault`	Layer 1	18 raw federal source files, ~14,811 observations; reports and triage records
`IAMGODIAM/bdi-sovereign-dataset`	Layer 2	1,574 verified observations, 8 pillars; quantitative specification
`IAMGODIAM/farmblock-data`	Layer 3, tract	15,507 tracts, 49 cities
`IAMGODIAM/farmblock-dataset`	Layer 3, county	24-county published pilot
`IAMGODIAM/bdi-black-paper`	Layer 4	This manuscript, drafts, peer review

Related campaign record: the DC Package — nine policy packets delivered to congressional leaders and to Justice Clarence Thomas in July 2026, comprising a 166-page shared evidentiary record per packet — is published at `dc.e5enclave.com`. This paper is the empirical instrument underlying that record.

APPENDIX G — ACKNOWLEDGMENTS AND INTELLECTUAL LINEAGE

W.E.B. Du Bois · Ida B. Wells · Harriet Tubman · Frederick Douglass · Nat Turner · Ella Baker · Malcolm X · James Baldwin · Cornel West · Melina Abdullah · The Movement for Black Lives

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Nil satis nisi optimum.

APPENDIX H — CORRECTIONS LEDGER

A. ERRORS FOUND BY RECOMPUTATION (not previously flagged)

These are new. They were not in the April triage matrix; they surfaced when the drafts' arithmetic was rebuilt from the source series.

#	DRAFT CLAIM	STATUS	VERIFIED VALUE	PRINT-EDITION LANGUAGE
R-1	"Black unemployment has never fallen below 2x the white rate" (Conclusion v2, v2.1)	**FALSE**	Ratio fell below 2.0 in **17 of 54 years**; low 1.576 (2020)	"The ratio has ranged from 1.58 to 2.56 and averaged 2.115 across 54 years. It has never inverted — not in one month of one year."
R-2	"Below 1.86, no normal economic year has gone in more than half a century" (Report 3, Adj. 3)	**FALSE**	2023 = 1.683, 2024 = 1.666 — both normal expansion years	Floor language removed. Replaced with the distributional statement above.
R-3	"3,015 Black Americans killed by police 2013–2023; 377 unarmed"	**UNDERCOUNT**	Vault row sum: **3,496 killed; 397 unarmed**	Corrected to 3,496 and 397; aggregate Black share 26.7% of 13,096 total.
R-4	"Incarceration ratio has never fallen below 5.7"	**FALSE**	1991–2022 window minimum **5.80 (2018)**; full 1925–2022 series minimum 5.44 (1940); maximum 7.70 (2000)	"In the empirical window the ratio has not fallen below 5.8. Across the full 97-year series it has ranged 5.44 to 7.70."
R-5	"Homeownership gap stands at 30.1 points today"	**WRONG VINTAGE**	ACS 2022: **28.9 pp**; ACS 2023 (NAR): 44.7% vs 72.4% = **27.7 pp**	"28.9 percentage points (ACS 2022); 27.7 points on the 2023 release."
R-6	"Wealth ratio reaches parity in approximately 263 years"	**ARITHMETIC ERROR**, then **WITHDRAWN ENTIRELY** (see W-1)	The 263-year figure was miscalculated; the recomputed 868-year figure rested on a defective series; the corrected series yields 274 years from 1989 but 1,632 from 1992 and no convergence from 2001	No parity horizon is asserted. The sensitivity table is printed in §5.1 instead.

#	DRAFT CLAIM	STATUS	VERIFIED VALUE	PRINT-EDITION LANGUAGE
R-7	"The gap tripled in absolute dollars"	**IMPRECISE**, then **SUPERSEDED** (see W-1)	Nominal 2.893×; in constant 2022 dollars **1.55×**, \$154,830 → \$240,120	"The absolute gap widened by \$85,290 in real terms and is the widest in the series."
R-8	"COVID erased 20 years of life expectancy gains in 18 months"	**TIMELINE WRONG**	Black life expectancy fell **4.0 years** across **two calendar years** (74.8 → 70.8, 2019→2021), returning to a mid-1990s level	"Between 2019 and 2021 — two years, not eighteen months — Black life expectancy fell four full years, to a level last seen in the mid-1990s."
R-9	"NAEP gap... exists in 2022 as it did in 1992" (Conclusion)	**OVERSTATED**	G8 reading gap narrowed 29.6 → 24.5; G8 math 40 → 32; G4 reading 32 → 28	Movement acknowledged and quantified; convergence rate stated.
R-10	"Eviction rate three times the white rate"	**SCOPE UNSTATED**	Eviction Lab 2000–2016: 3.00–3.54×. Graetz et al. 2023 PNAS: ~4× evictions, ~4.8× filings	Both cited with explicit scope and instrument.

B. TRIAGE MATRIX RULINGS APPLIED (April 18, 2026 — 27 claims)

ID	RULING	PRINT-EDITION DISPOSITION
E-1	source-confirmed → **REVERSED** (see W-1)	The 1989 endpoint could not be reconciled to the Federal Reserve's published race-specific medians under either price basis. Table rebuilt from the Fed's constant-2022-dollar series.
E-2	source-confirmed	~30 pp restated as 28.9 pp (ACS 2022) per R-5.
E-3	source-conflicted	"Ranged 2.24 to 2.84 across ACS 2005–2022" — no "never below 2:1" absolute.
E-4	source-confirmed	1968 ≈ 24 pp vs 2022 28.9 pp. Both vintages footnoted.
H-1	source-confirmed	69.9/100K locked to **2021**; 2022 = 49.5 shown; NCHS 2024 trajectory noted.
H-2	source-conflicted	Rewritten per R-8.
H-3	source-conflicted	Fence-line tract geography specified: 65–94% Black (St. James Districts 4–5); parish ≈44%.
CJ-1	source-conflicted	Rewritten per R-4.
ED-1	source-conflicted	24-point gap and century-plus parity horizon — **independently confirmed against the live NAEP API** (see §C).
HO-1	source-confirmed	HMDA denial ratio; range 2.01–2.36 stated rather than a single figure.
HO-2	source-conflicted	Rewritten per R-10.
HO-3	internally-derived	**\$4.5B aggregate removed.** Replaced with HUD OIG 2013-AT-0003: 53% of PHAs did not file required Section 3 reports.

ID	RULING	PRINT-EDITION DISPOSITION
HO-4	source-conflicted	"Erased a decade in 18 months" → "reversed nearly a decade of gains across the 2007–2013 window."
HI-1	source-conflicted	Embarked/disembarked disambiguated: 12,521,337 / 10,702,656 / 1,818,681 lost (14.5%).
HI-2	source-confirmed	1514 anchor retained with footnote.
EN-1	citation-vintage	EJScreen cited via Public Environmental Data Partners; EPA.gov removal (Feb 5, 2025) disclosed.
CD-1	source-confirmed	Humphreys 78.6% (ACS 2022).
CD-2	source-confirmed	Detroit 77.1%.
CD-3	source-confirmed	East St. Louis 97.4%.
CD-4	source-confirmed	Claiborne 86.1%.
CD-5	source-conflicted	Same fix as H-3.
CD-6	citation-vintage	**32.1% all-persons and 55.0% child poverty (ACS 2022)** adopted as the citation figures; 36.4% (ACS 2015–2019) retained as the historical vintage; ACS 2024 5-year now reports 27% ±6.3 — disclosed as the current release.
CD-7	source-confirmed	Hospital closure 2013 (HRSA AHRF).
CD-8	source-conflicted	"No grocery within 15 miles" → USDA FARA low-income/low-access (LILA) designation.
CD-9	internally-derived → **published**	Humphreys score corrected **83.5 → 87.25** and the full five-component decomposition printed as Appendix E.
META-1	source-confirmed	**1,855 → 1,574** verified empirical observations, everywhere.
META-2	internally-derived	"MiroFish 94%" removed. Replaced with the dual-source protocol description.

B2. SECOND-WAVE CORRECTIONS — INDEPENDENT REVIEW (v1.2)

On September 1, 2026 an independent review of the v1.1 print edition was conducted by Manus AI, working only from the published PDF, the corrections ledger and the supplied git bundle — with no access to this project's repositories. It found problems the internal audit had missed, and its principal finding is the most material error yet identified in this paper. The findings are adopted below.

Why the internal audit missed them. The recomputation pass described in §A verified that every derived figure followed correctly from the vault's stored series. It did. What it never asked was whether a stored series was itself coherent, or whether it reconciled to the agency's published figures. An internal consistency check cannot detect an incoherent input. The audit protocol is amended: **primary-source reconciliation of every series endpoint is now a required pass.**

#	CLAIM	FINDING	PRINT-EDITION DISPOSITION
W-1	**Table 5.1, the SCF wealth series; the "$83,000 → $240,100" gap; "nearly tripled"; the 3.2-cent ratio gain; the 868-year parity horizon**	**CRITICAL — series defective.** The table stated no price basis. Its 1989 endpoint ($12k / $95k) was nominal while its 2022 endpoint ($44.9k / $285k) was the Fed's constant-2022-dollar figure — a real endpoint back-filled with nominal history, whose apparent trend was substantially an inflation artifact. Worse, deflating the old 1989 values to 2022 dollars yields $28,321 and $224,211 against the Fed's published $9,200 and $164,030 — so they reconcile under **neither** basis, and their provenance was undocumented.	**Table rebuilt in full** from Federal Reserve FEDS Notes (Oct 18, 2023), Figure 2, constant 2022 dollars, families. Corrected findings: real gap **$154,830 → $240,120** (1.55×, +$85,290, widest in the series); ratio **0.056 → 0.158** (nearly tripled); Black median wealth **+388%** vs white **+74%**. The "nearly tripled gap," the 3.2-cent gain and the 868-year horizon are all withdrawn.
W-2	Parity projections (wealth and NAEP)	Presented with insufficient warning that they are mechanical extrapolations. The wealth horizon proves extremely baseline-sensitive: 274 years from 1989, 1,632 from 1992, 1,511 from 1995, 366 from 2007, 117 from 2013, and **no convergence at all** from 2001.	**No wealth parity horizon is asserted.** The full sensitivity table is printed in §5.1. The NAEP 144-year figure is retained but explicitly labelled an arithmetic extension of two endpoints, not a forecast.
W-3	Incarceration ratios, and "Black Americans were imprisoned at 6.31 times the white rate"	**Definition unverified and probably overstated in scope.** BJS publishes, for 2022, **1,196 per 100,000 Black adult residents and 229 per 100,000 white adult residents — a ratio of 5.22.** The series' 1,862 and 295 correspond closely to BJS *male* rates (≈1,826 and 279). The vault does not record the denominator, and the Cahalan (1986) pre-1980 crosswalk is undocumented. BJS also dates its series to 1926, not 1925.	Chapter 6 now prints the BJS all-adults ratio of **5.22** alongside the series ratio of 6.31, states that the basis is probably male rates, and stops writing "Black Americans" where the source may say "Black males." The stability finding, which holds on either basis, carries the argument.
W-4	1930 maternal mortality ratio of 1.48, and the 1.76-fold widening	The pre-1933 figures come from the birth-registration states only, which were not nationally representative. The cross-era comparison is not strictly commensurable.	Caveat added in full. The 2022 ratio is confirmed directly against NCHS (49.5 ÷ 19.0 = 2.605 → 2.61). The chapter now notes that the direction holds across every intermediate decade, and offers the 2010→2022 rise (2.24 → 2.61), entirely within a fully registered system, to readers who reject the 1930 datum.
W-5	Unemployment ratios shown beside rounded annual rates	5.5 ÷ 3.3 = 1.667, not the 1.683 printed. The discrepancy arises because ratios are computed from unrounded monthly data while the displayed component rates are rounded annual averages.	Aggregation rule now stated explicitly in §5.2, with the worked 2023 example (5.5167 ÷ 3.2750 = 1.685) and a vintage-revision note. The stored values are correct; the method was undisclosed.

#	CLAIM	FINDING	PRINT-EDITION DISPOSITION
W-6	Black farmland "13.5 million acres lost, 1910–1997"	The 15-million-acre peak's year and level are contested; some compilations place the maximum nearer 1900.	Note added. The 1997 figure of 1.5 million is well supported; the 90 percent loss is now framed against a peak carrying acknowledged uncertainty in year and level.
W-7	NAEP endpoint precision and comparability	NCES publishes rounded scores (267/237, 268/244) and discloses that 1992 permitted no testing accommodations while later years did.	Both disclosed in §6.6.
W-8	The supplied git bundle	**Unusable for independent review.** It was created incrementally (`main..branch`) and declares a prerequisite commit the reviewer did not have, so the repository tree, raw vault JSON and manifests could not be inspected at all.	Replaced with a **complete-history bundle** requiring no prerequisite. This was the single largest obstacle to the review and it was an avoidable packaging error.
W-9	Introduction's "Five Compound Catastrophe Zones" heading	Flagged as an apparent contradiction with the Chapter 7 ranking.	**Already corrected before the review was received** — the reviewer worked from the superseded v1.1 PDF. The Introduction now names the instrument's actual top five and states the withdrawal.

Findings acknowledged and not yet closed. The review's three "critical" reproducibility actions are only partly satisfied by this edition. The complete bundle (W-8) addresses repository access. It does **not** yet supply a full data package for the FDI computation — raw inputs, normalization bounds, transformation code and a deterministic rebuild command — nor ACS table identifiers and geography codes for every Humphreys input. Until those exist, the correct characterisation is the reviewer's: **the FDI outputs are not independently reproducible from the supplied package.** Appendix E says so.

C. LIVE FEDERAL-SOURCE VERIFICATION (August 2026)

SERIES	ENDPOINT	RESULT
Black unemployment (LNS14000006)	BLS via FRED, monthly 2019–2025	Annual averages reproduce the vault series. 2019 = 6.1, 2023 = 5.5, 2024 = 5.9, 2025 = 6.9. **Confirms vault.**
White unemployment (LNS14000003)	BLS via FRED, monthly 2019–2025	2019 = 3.3, 2023 = 3.3, 2024 = 3.6, 2025 = 3.7. **Confirms vault.** Ratios recomputed from live monthly data reproduce R-1 and R-2.
NAEP G8 reading by race, national	NCES Nation's Report Card data service	**1992:** White 267.001, Black 237.374 → gap **29.63**. **2022:** White 268.443, Black 243.902 → gap **24.54**. Confirms ED-1 exactly: the gap is ~24 points, not 20. Convergence 5.09 points in 30 years → **~144 years to parity**, not 64.

SERIES	ENDPOINT	RESULT
Humphreys County, MS (FIPS 28053)	Census ACS 5-year profile	Current release reports 27% ±6.3 poverty, median household income $33,731, population 7,395. Wide margin of error at this population size disclosed in text.
Homeownership by race	ACS 2023 via NAR *Snapshot of Race and Home Buying in America* (2025)	Black 44.7%, White 72.4% → 27.7 pp. Confirms direction; updates the vintage.

Note on API access. The Census Bureau data API now requires a registered key for programmatic access; the figures above were verified against the Bureau's published profiles and released tables rather than by unauthenticated API call. The vault's underlying ACS pulls (committed April 2026) remain the citation of record for the series tables, and each is footnoted to its table number.

D. STRUCTURAL ADDITIONS

ITEM	ACTION
Chapters 5, 6, 7, 8	**Drafted** from the locked outline; previously stubs.
Introduction	**Drafted**; previously a stub.
Appendices A–H	**Drafted**; previously stubs.
Report 3 drafting directives 3–8	Executed inside the new chapters (unemployment distribution, maternal mortality reversal, COVID reversal, housing headline, Shelby inflection, three historical anchors).
Scope statement (Adj. 1)	Introduction now states the nested window: 1991–2024 core, with series to 1900 (health), 1925 (justice), 1940 (housing), 1964 (political), 1514 (historical).
Chapter 2 longitudinal claim (Adj. 2)	Added.

E. WHAT A REVIEWER SHOULD STILL CONTEST

Disclosed, not concealed. A sovereign record that hides its soft edges is promotional, not empirical.

1. **The `% Black` variable at 15% weight in the county FDI.** Defended as a structural-exposure proxy in Appendix E; a reviewer may reasonably argue it makes the instrument partly tautological with respect to racial concentration. The tract instrument excludes it — compare the two.
2. **Equal weighting in the tract FDI.** A transparent baseline, not an empirically derived one. PCA weighting is the stated v3.0 path.
3. **The Humphreys decomposition reproduces 79.7, not 87.25**, when recalculated against approximate corpus bounds. The published score stands on the full-corpus normalization; the arithmetic gap is printed in Appendix E rather than smoothed over.

4. **D4 health imputed to zero** where CDC PLACES is unavailable — conservative, but it understates distress in data-sparse tracts.
5. **The food-desert proxy** substitutes an ACS income/poverty threshold for a FARA download that 404'd at collection. Rows are flagged.
6. **Mapping Police Violence is not a federal instrument.** It is the most complete public database of its kind, and it is labeled as non-federal wherever cited.
7. **Small-county ACS margins of error** — Humphreys' poverty estimate carries a ±6.3-point margin. Stated in text.
8. **The incarceration series' denominator is not established** (W-3). It is probably male rates; BJS's all-adults 2022 ratio is 5.22 against the series' 6.31. Both are printed. The crosswalk is owed.
9. **The FDI computation is not independently reproducible** from the published package (see Appendix E.4a). The reviewer's characterisation is adopted verbatim.
10. **The pre-1933 maternal mortality figures** rest on the birth-registration states only and are not nationally representative (W-4).
11. **Table 5.1 now depends on a single secondary presentation** of the SCF — the Federal Reserve's own FEDS Note — rather than on an extraction from SCF microdata performed by this project. That is an improvement in provenance and a reduction in independence. A microdata extraction with published code is owed.
